

Nonclassicality, quasiprobabilities and weak values explored in neutron interferometry

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- Introduction
- Nonclassicality
- Quasiprobabilities and weak values
- Experimental results

Introduction

Quantum mechanics exhibits features with no classical counterpart.

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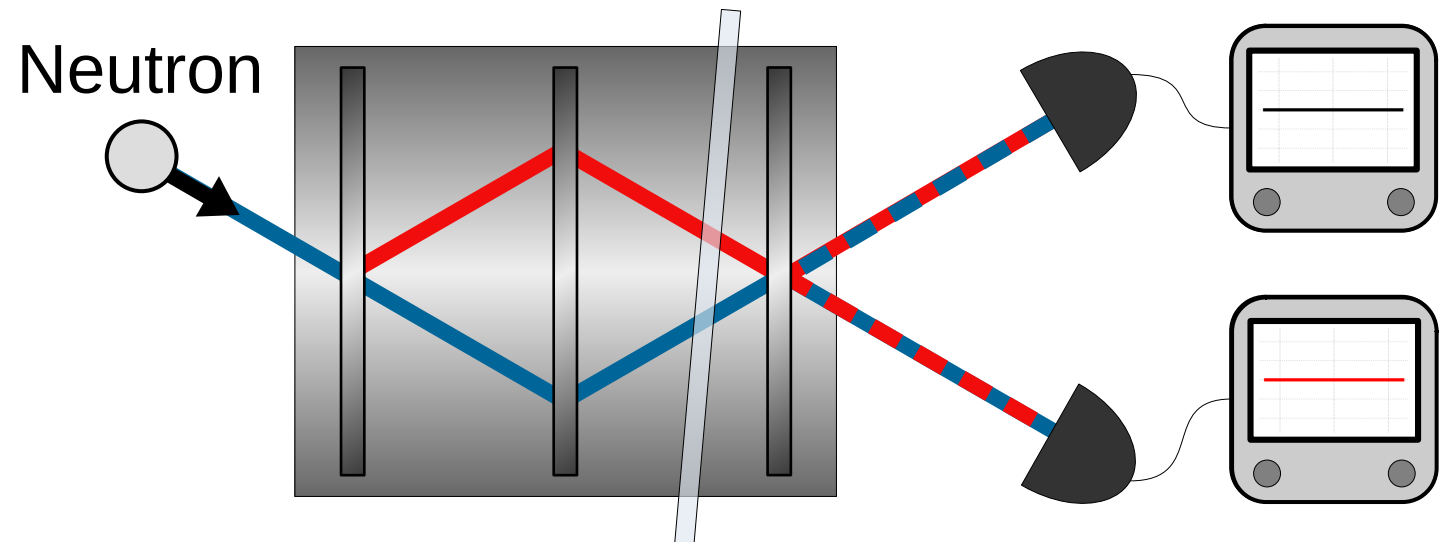
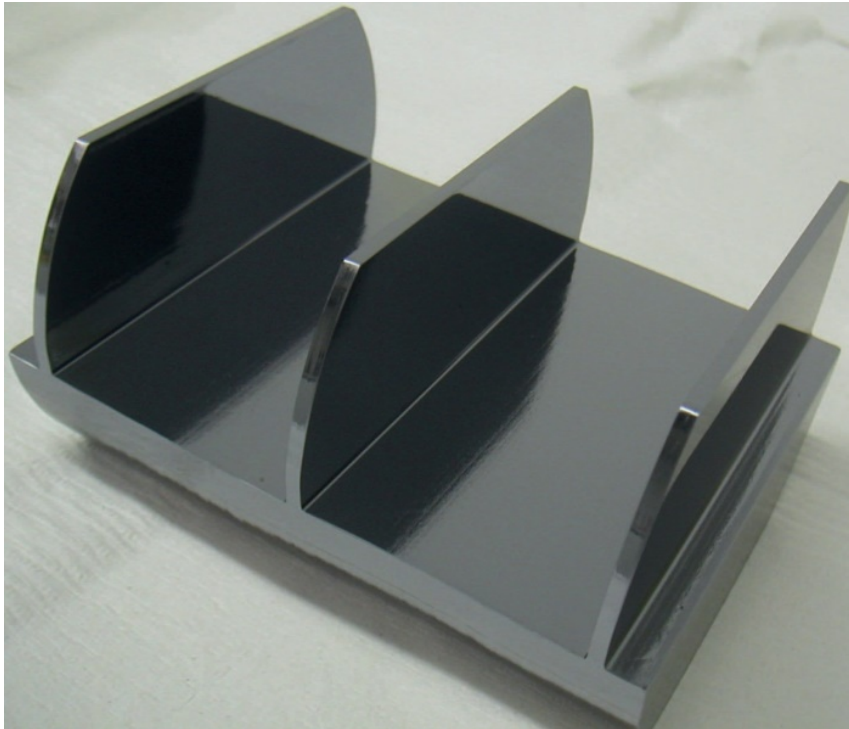
However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

Quantum mechanics exhibits features with no classical counterpart.

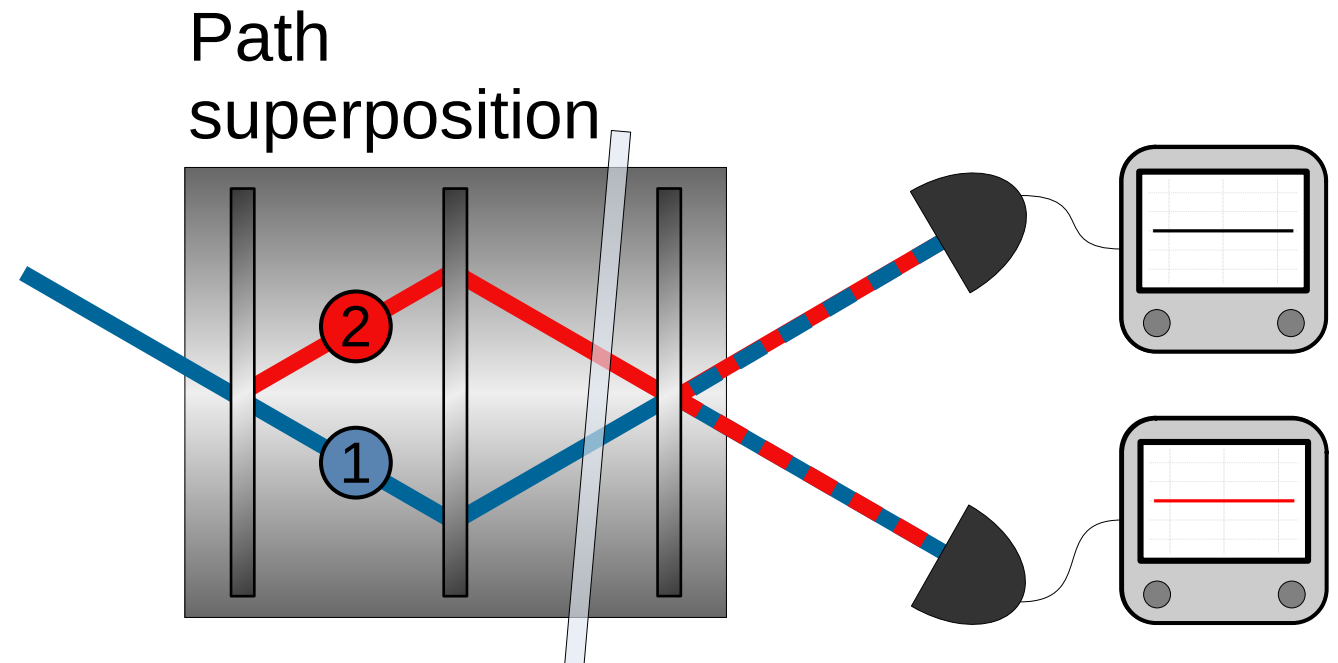
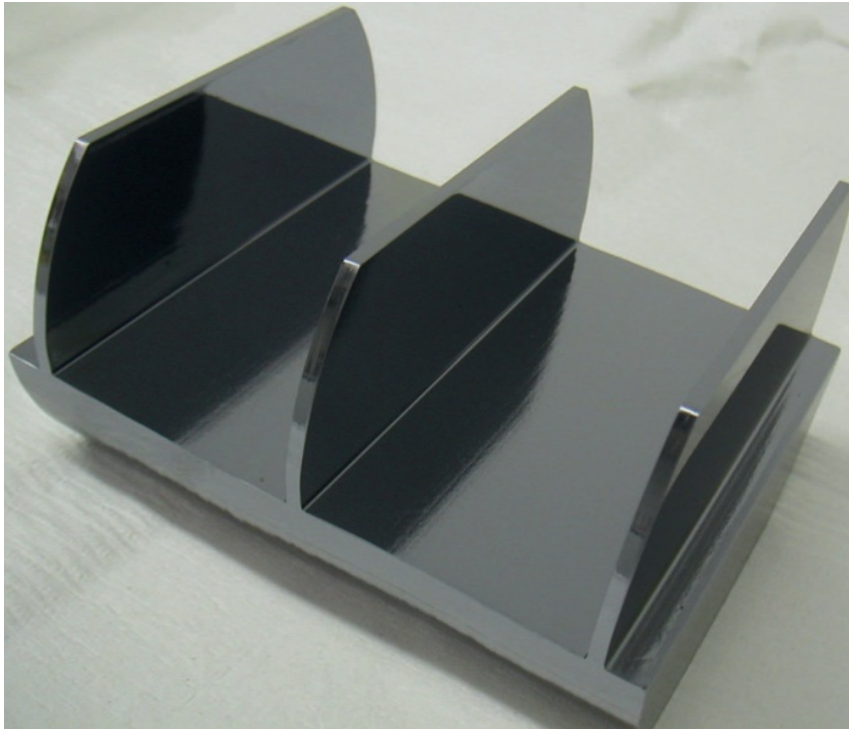
However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

Quasiprobability representations and weak values provide a valuable tool to probe such boundaries.

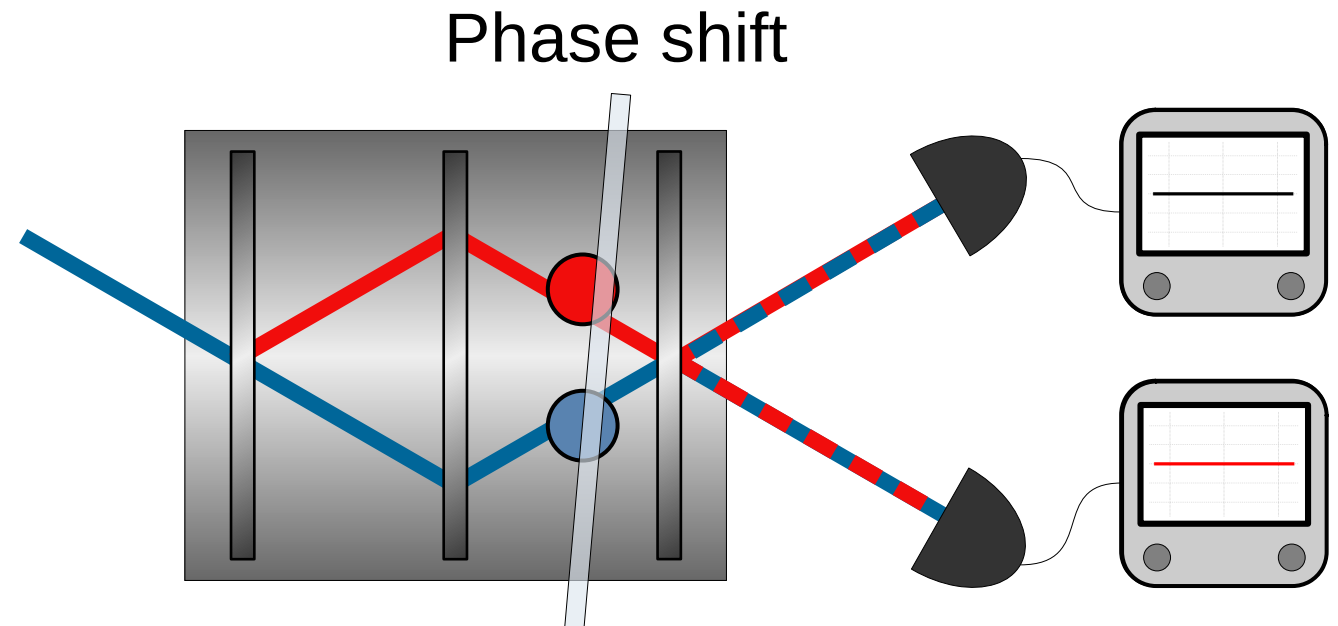
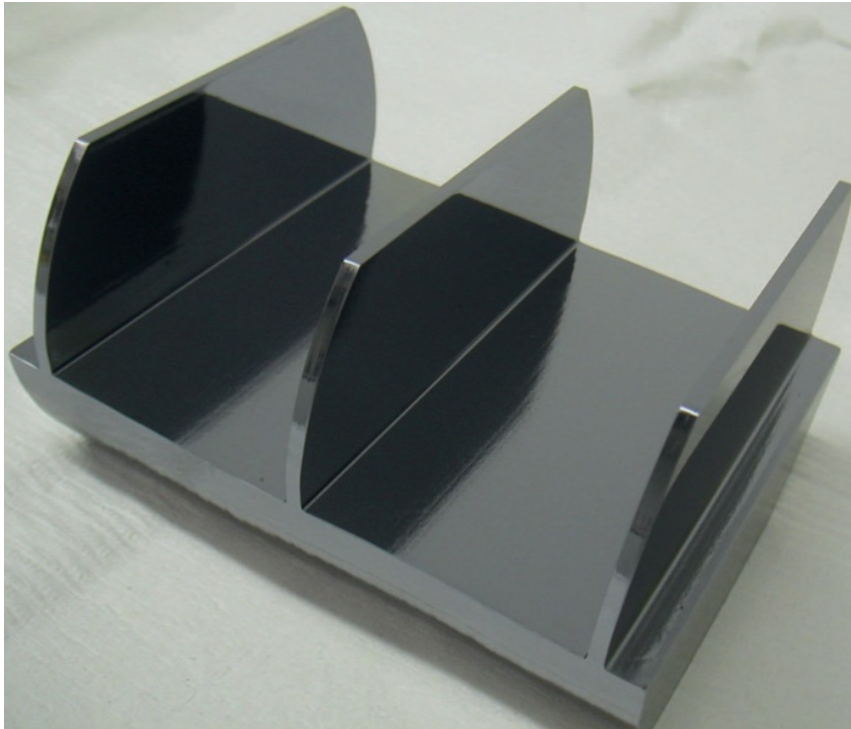
Neutron Mach-Zehnder interferometer:



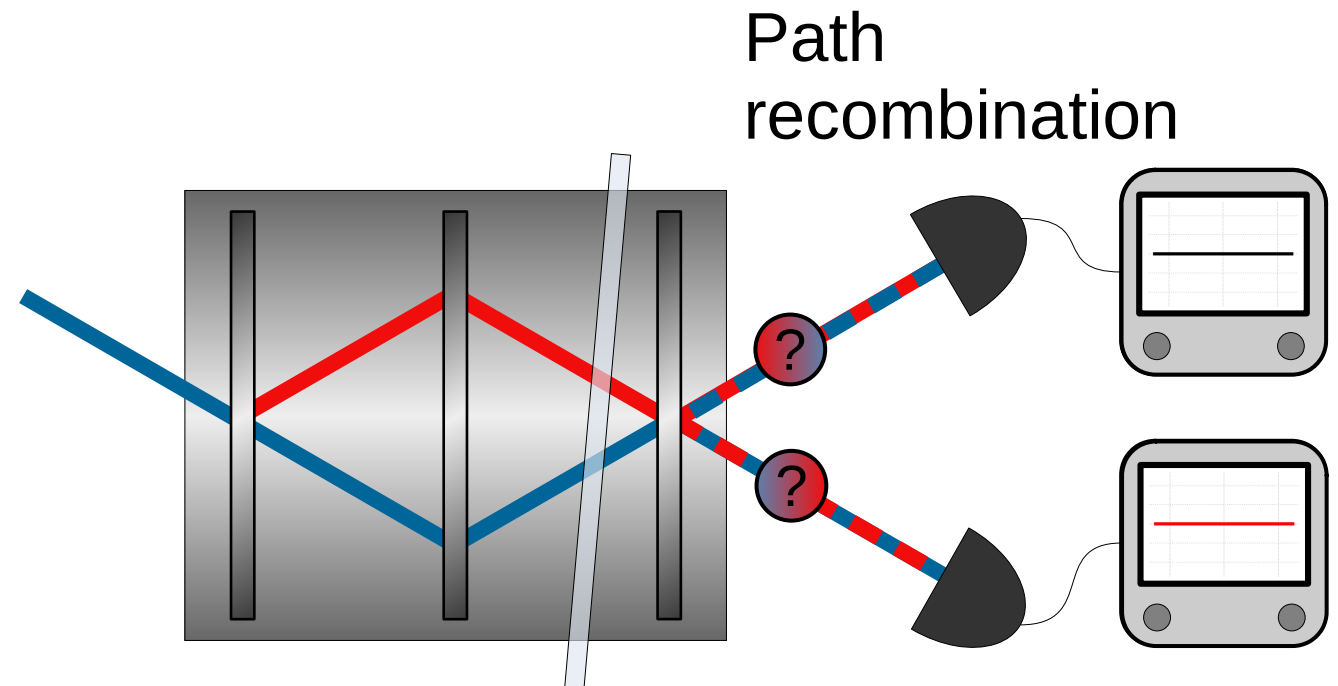
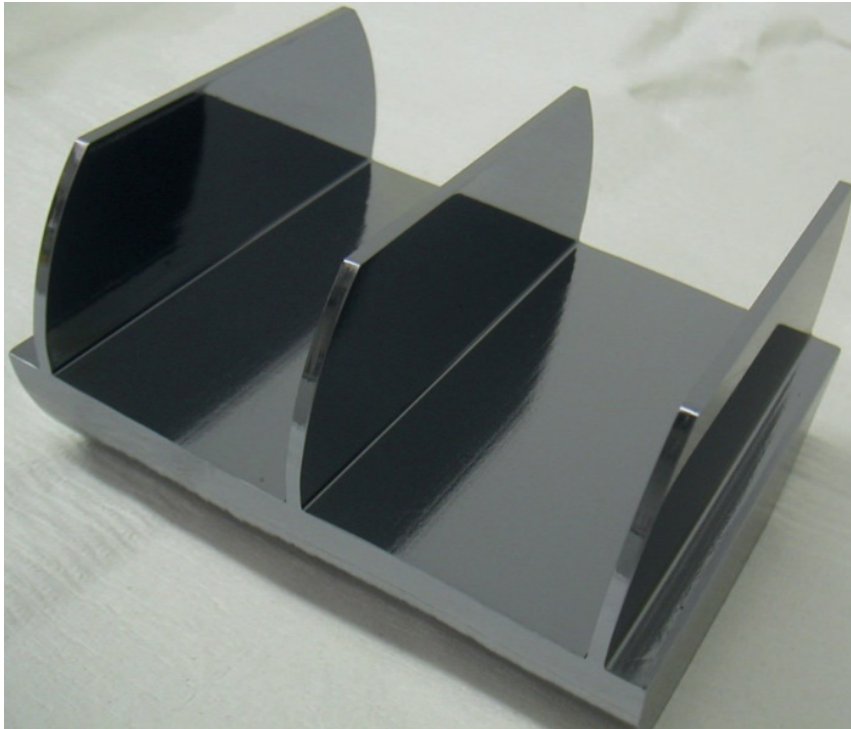
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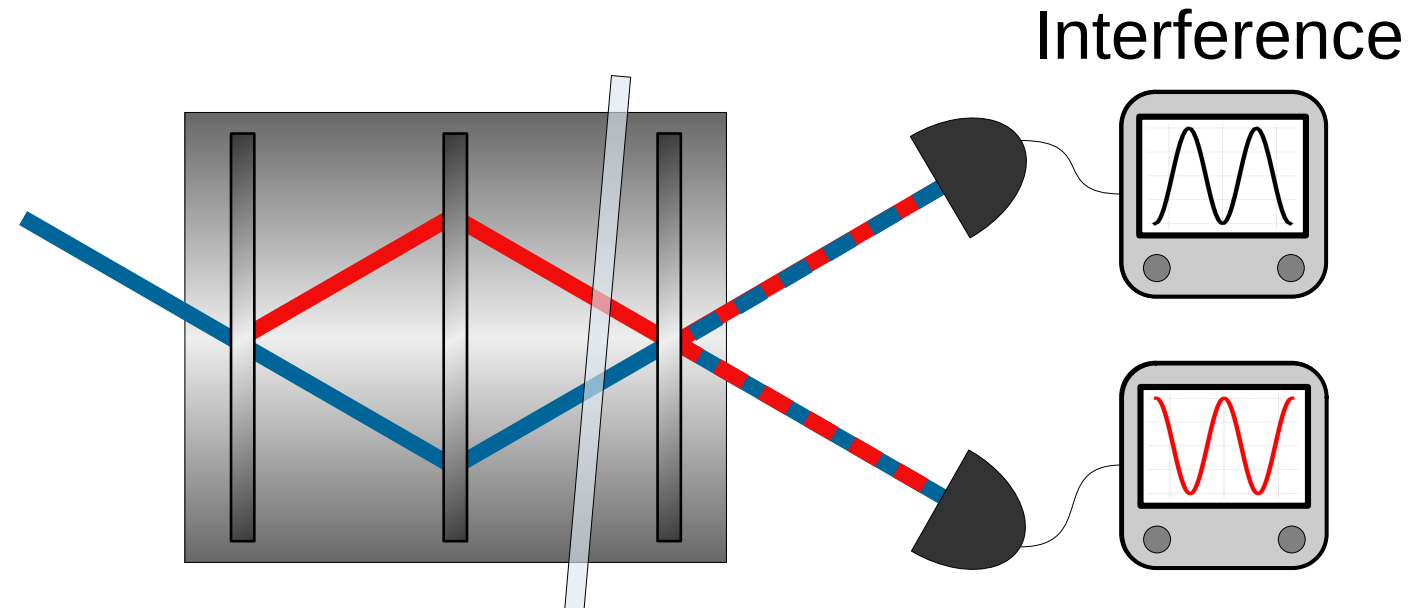
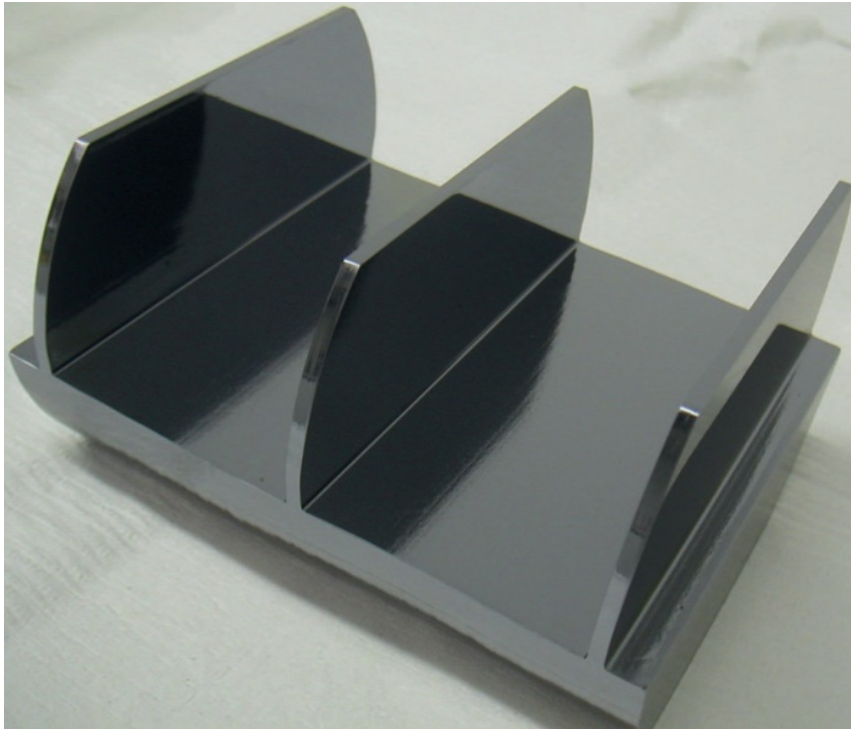
Neutron Mach-Zehnder interferometer:



Neutron Mach-Zehnder interferometer:



Neutron Mach-Zehnder interferometer:



Why neutron interferometry?

- Interference is intrinsically a single-massive-particle phenomenon.
- Matter-wave interferometry at room temperature and ambient pressure, with macroscopic beam separation and high contrast.
- Manipulation of several degrees of freedom (path, spin, energy).

Nonclassicality

Preparation (P)



Preparation (P)



System



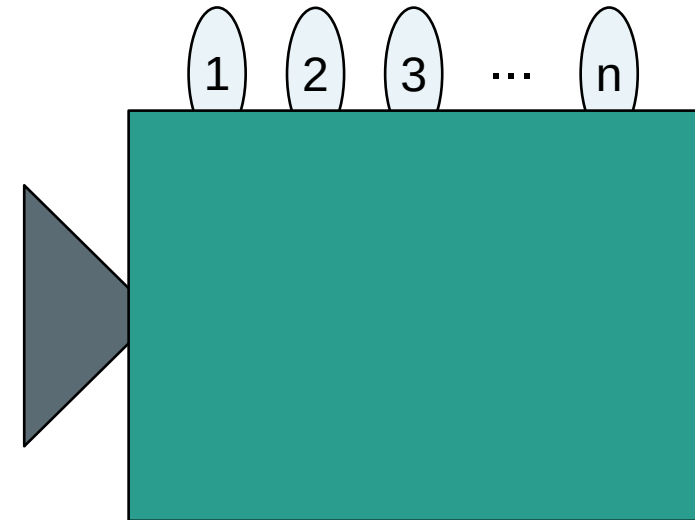
Preparation (P)



System



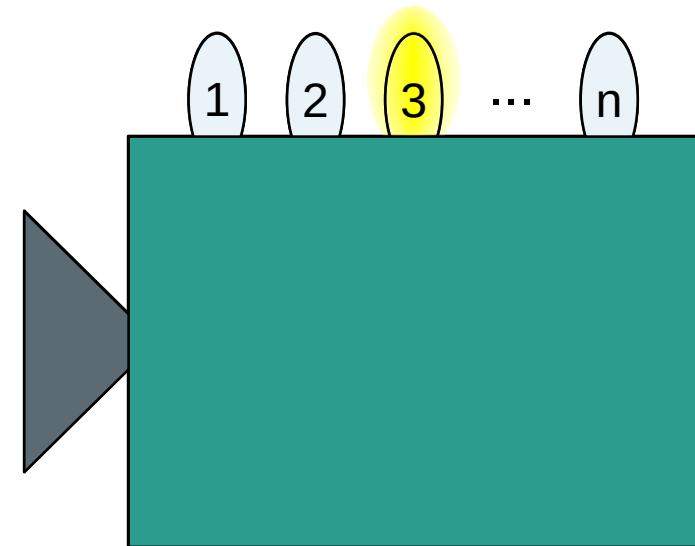
Measurement (M)



Preparation (P)



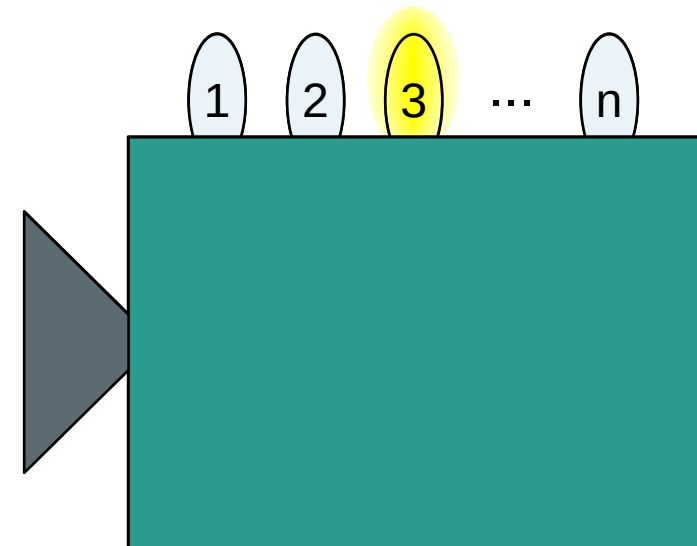
Measurement (M)



Preparation (P)

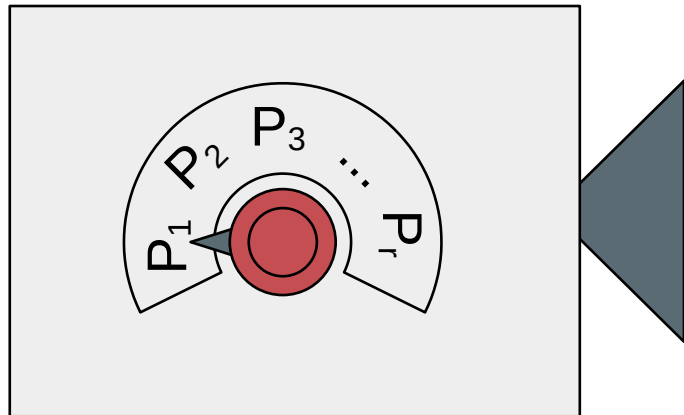


Measurement (M)

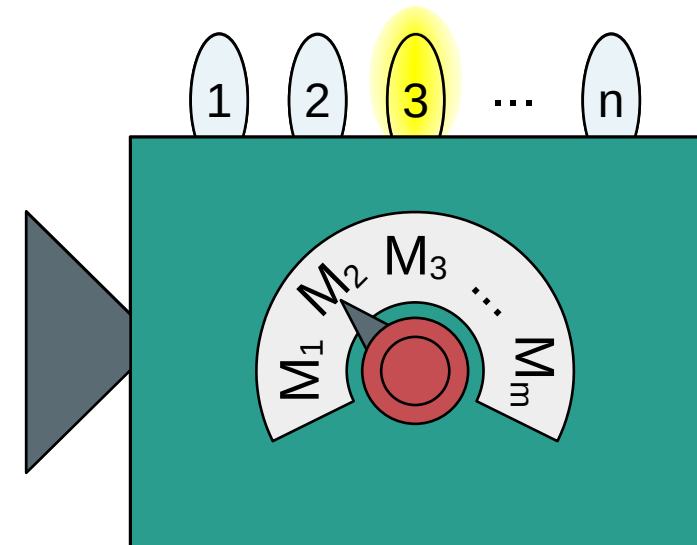


Outcome probabilities: $\text{Prob}(k|P, M)$

Preparation (P_i)

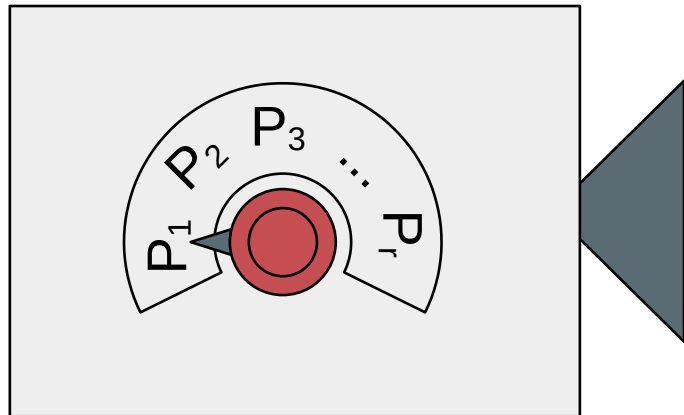


Measurement (M_j)



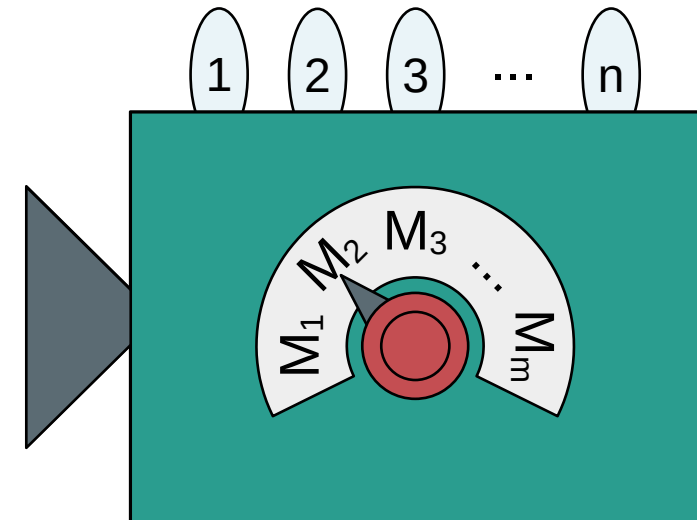
Outcome probabilities: $\text{Prob}(k|P_i, M_j)$

Preparation (P_i)



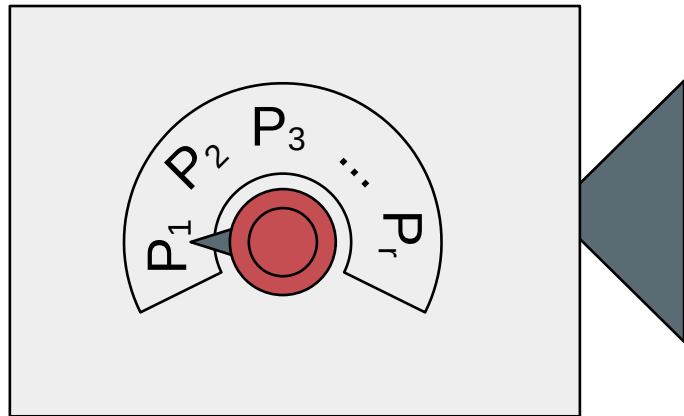
System
 λ

Measurement (M_j)

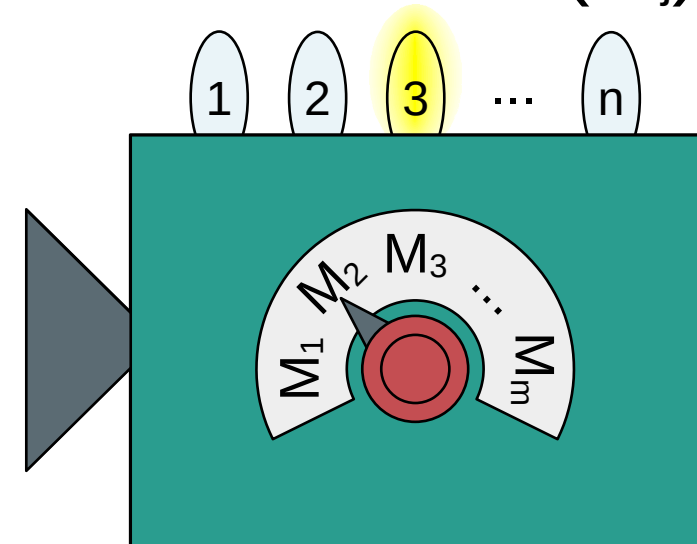


Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_m)$

Preparation (P_i)

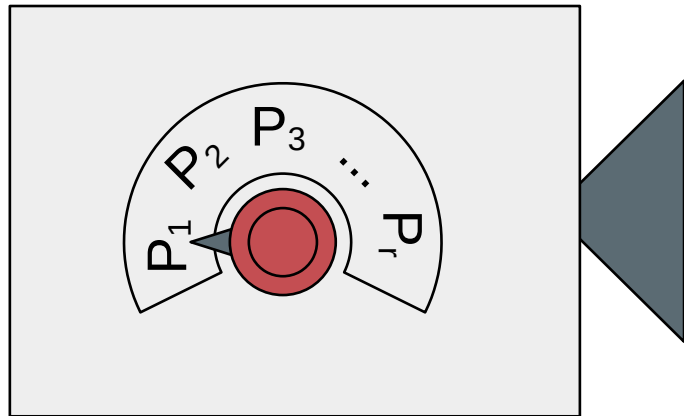


Measurement (M_j)

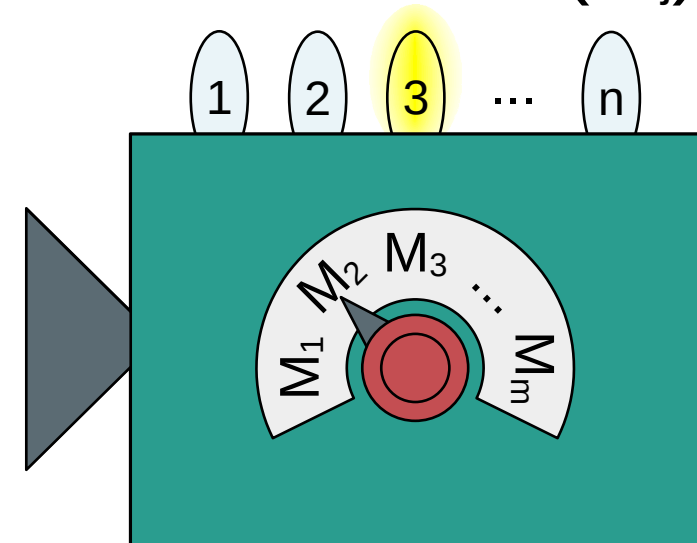


Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_m)$

Preparation (P_i)

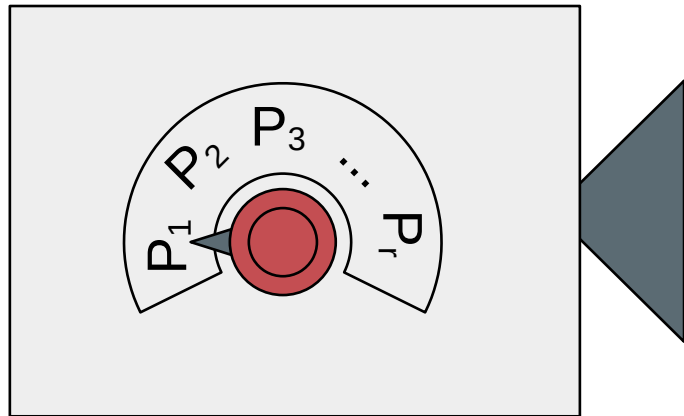


Measurement (M_j)

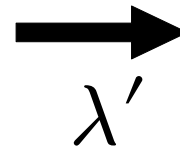


Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_m)$

Preparation (P_i)

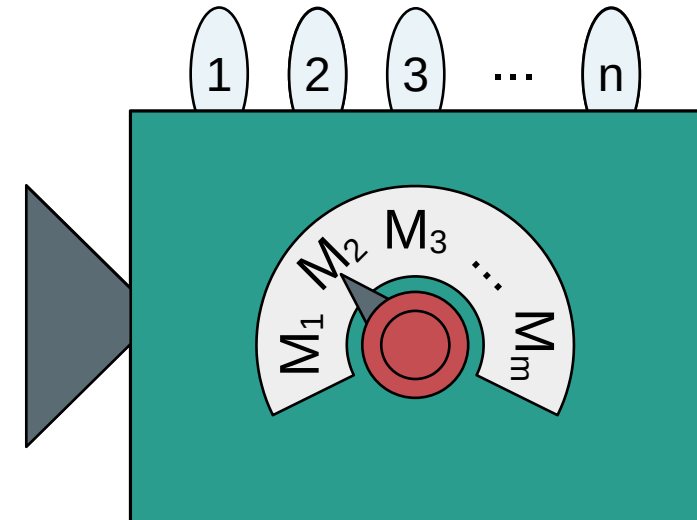


System

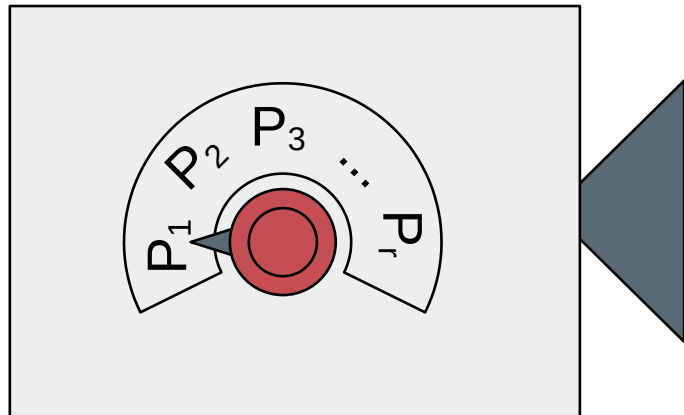


λ'
100%

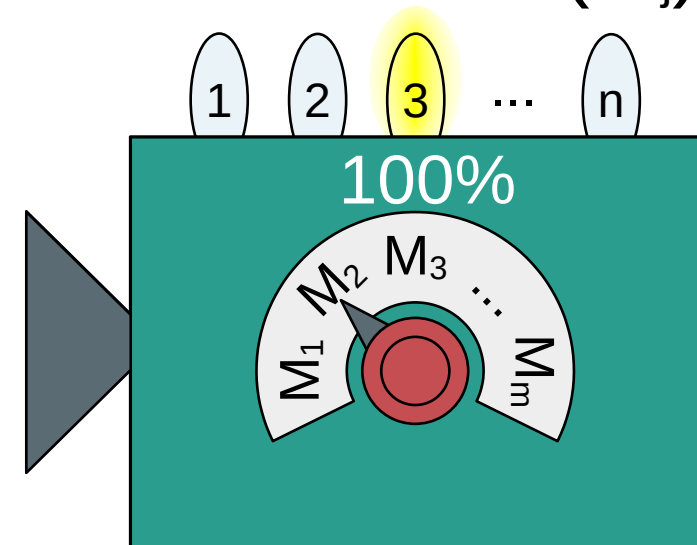
Measurement (M_j)



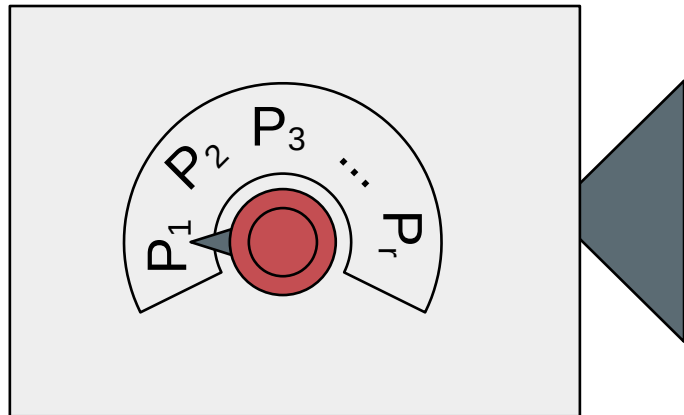
Preparation (P_i)



Measurement (M_j)

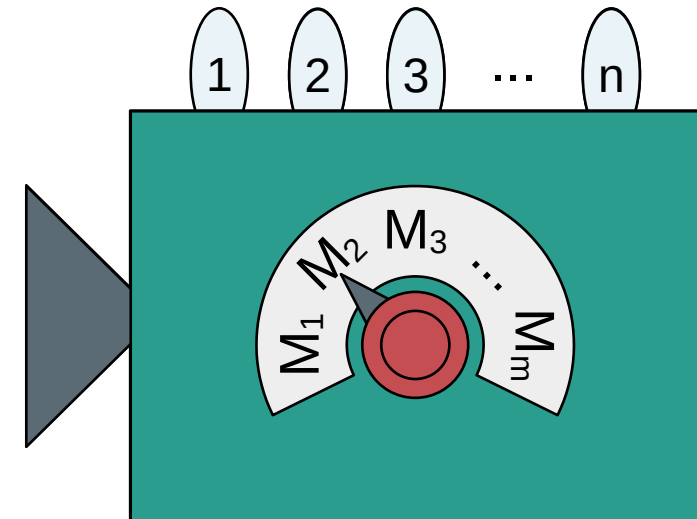


Preparation (P_i)



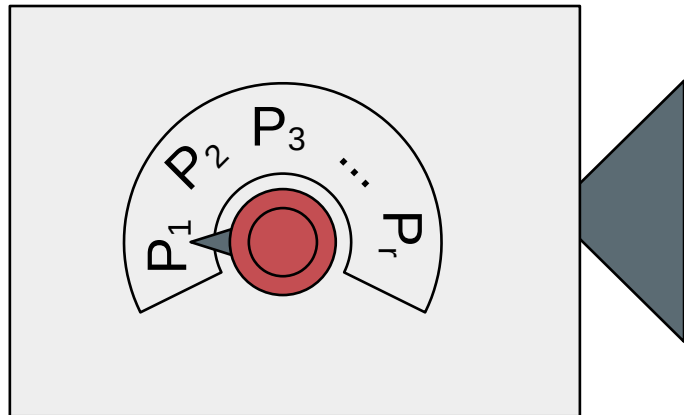
System
 λ' λ''
75% 25%

Measurement (M_j)



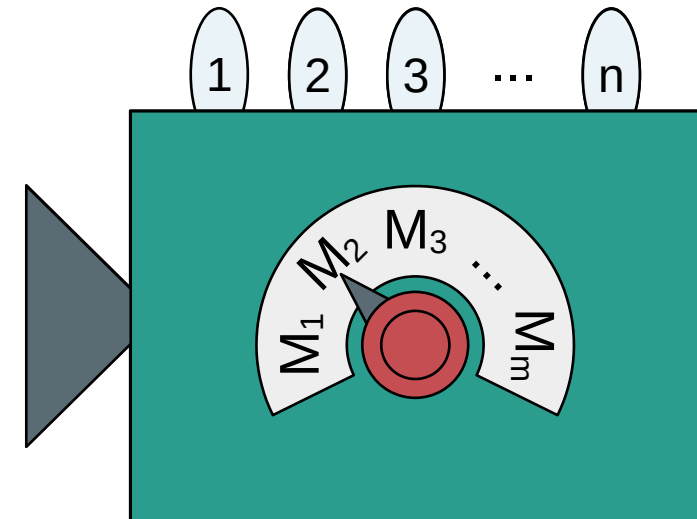
Probability distribution: $\mu(\lambda|P_i)$

Preparation (P_i)



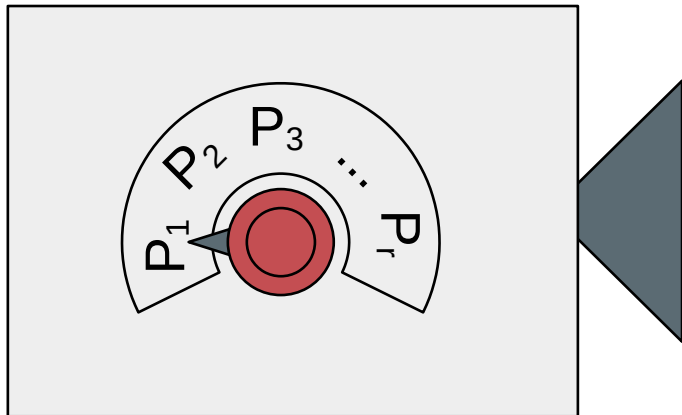
System
 λ' λ''
75% 25%

Measurement (M_j)

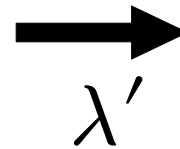


Probability distribution: $\mu(\lambda|P_i)$

Preparation (P_i)

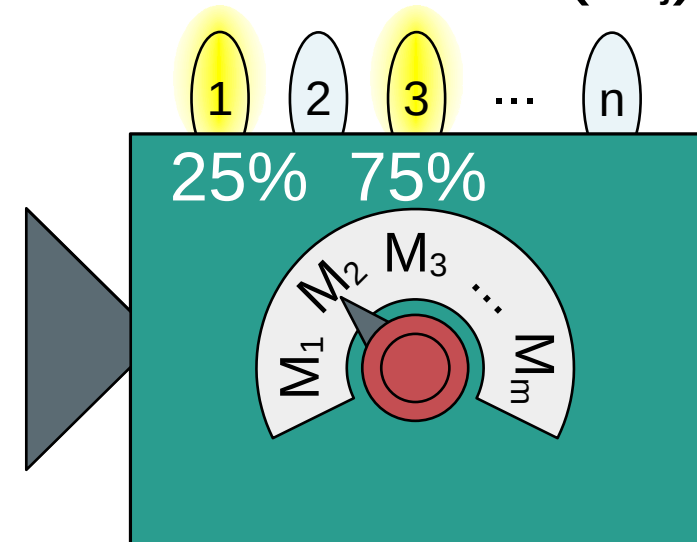


System



λ'
100%

Measurement (M_j)

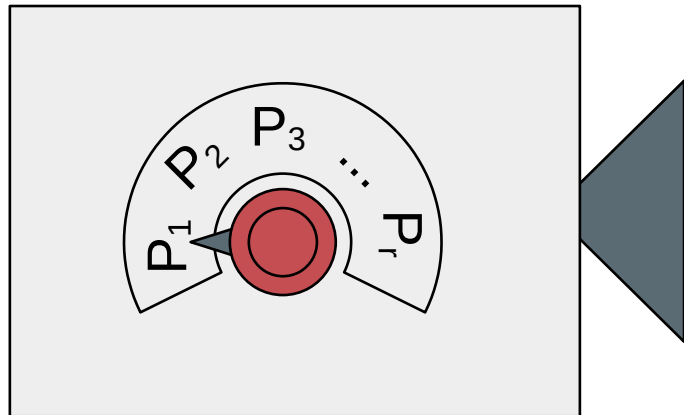


A classical experiment

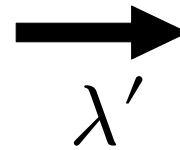
Probability distribution: $\mu(\lambda|P_i)$

Response function: $\xi(k|M_j, \lambda)$

Preparation (P_i)

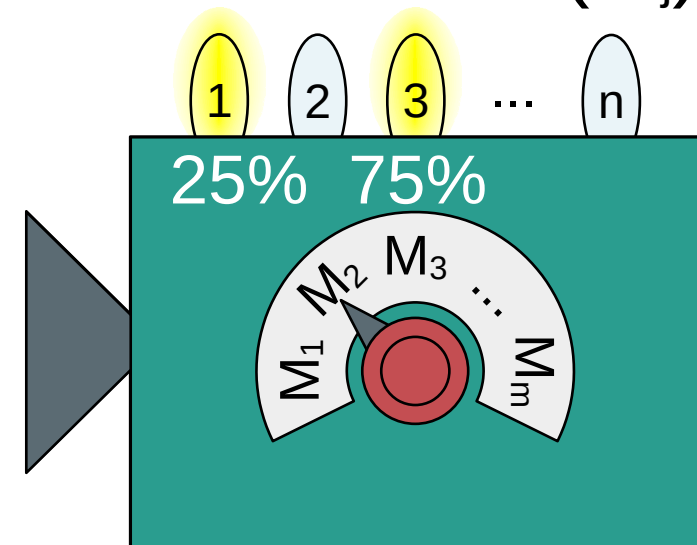


System



λ'
100%

Measurement (M_j)



Probability distribution: $\mu(\lambda|P_i)$ Response function: $\xi(k|M_j, \lambda)$

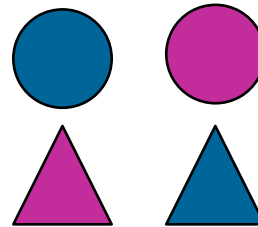
Outcome probabilities:

$$\text{Prob}(k|P_i, M_j) = \sum_{\lambda} \mu(\lambda|P_i) \xi(k|M_j, \lambda)$$

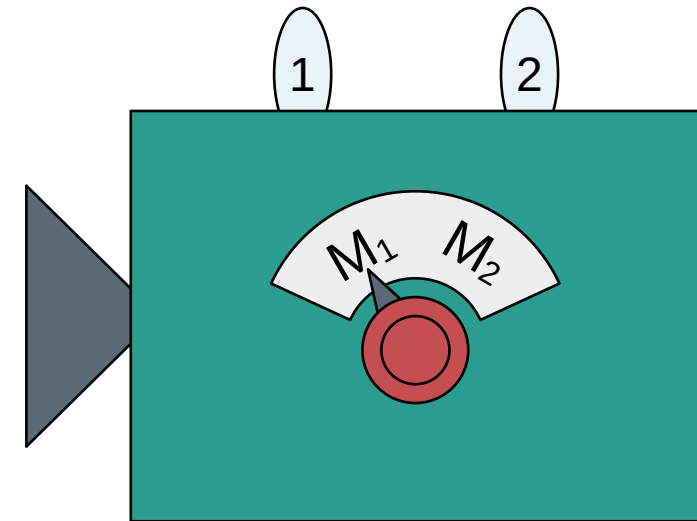
Preparation (P)



System



Measurement (M_j)



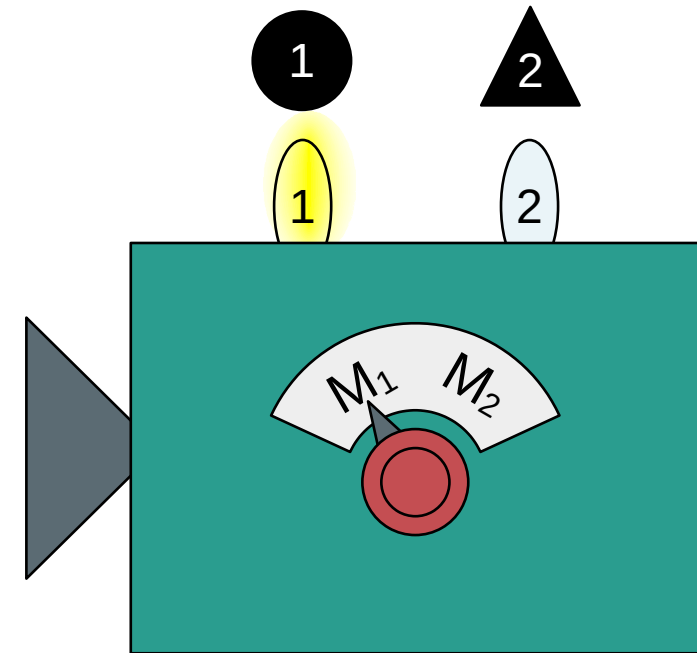
Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment

Preparation (P)



Shape (M_1)

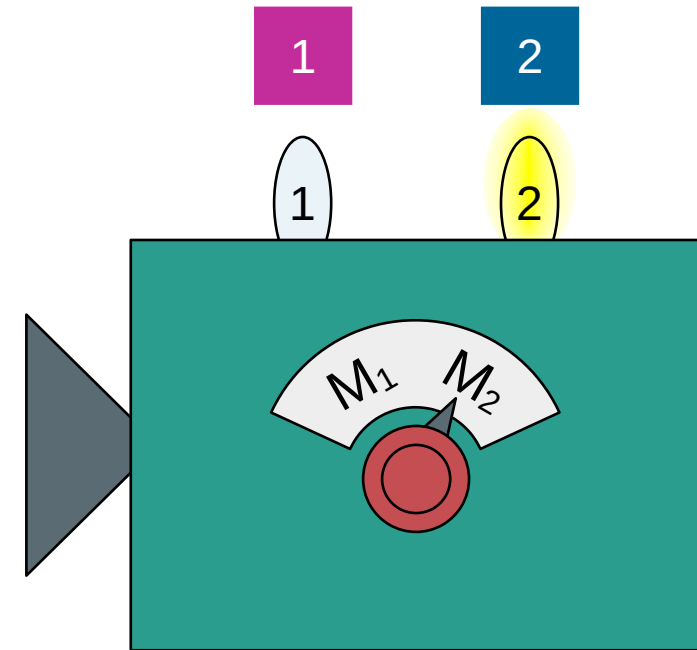


Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

Preparation (P)



Color (M_1)



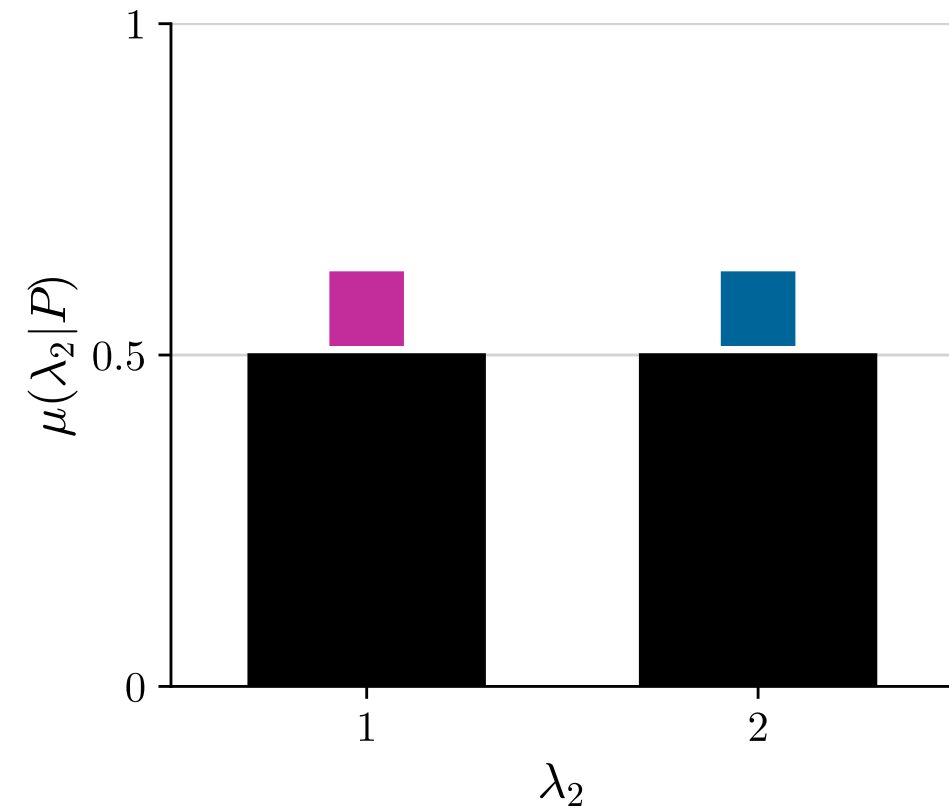
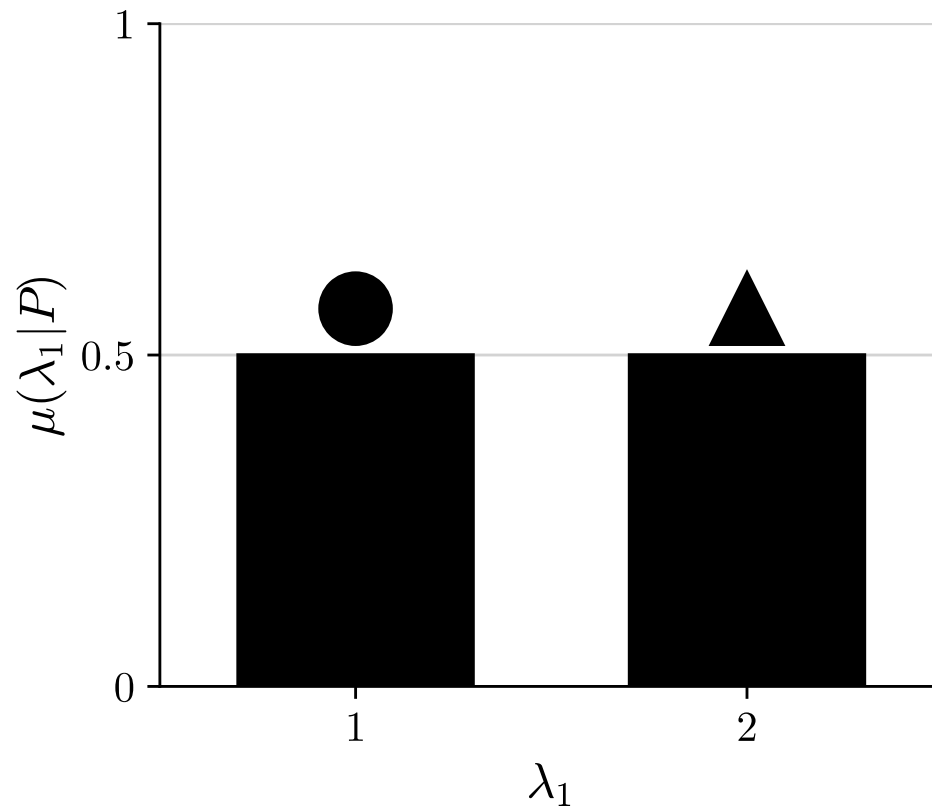
Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment

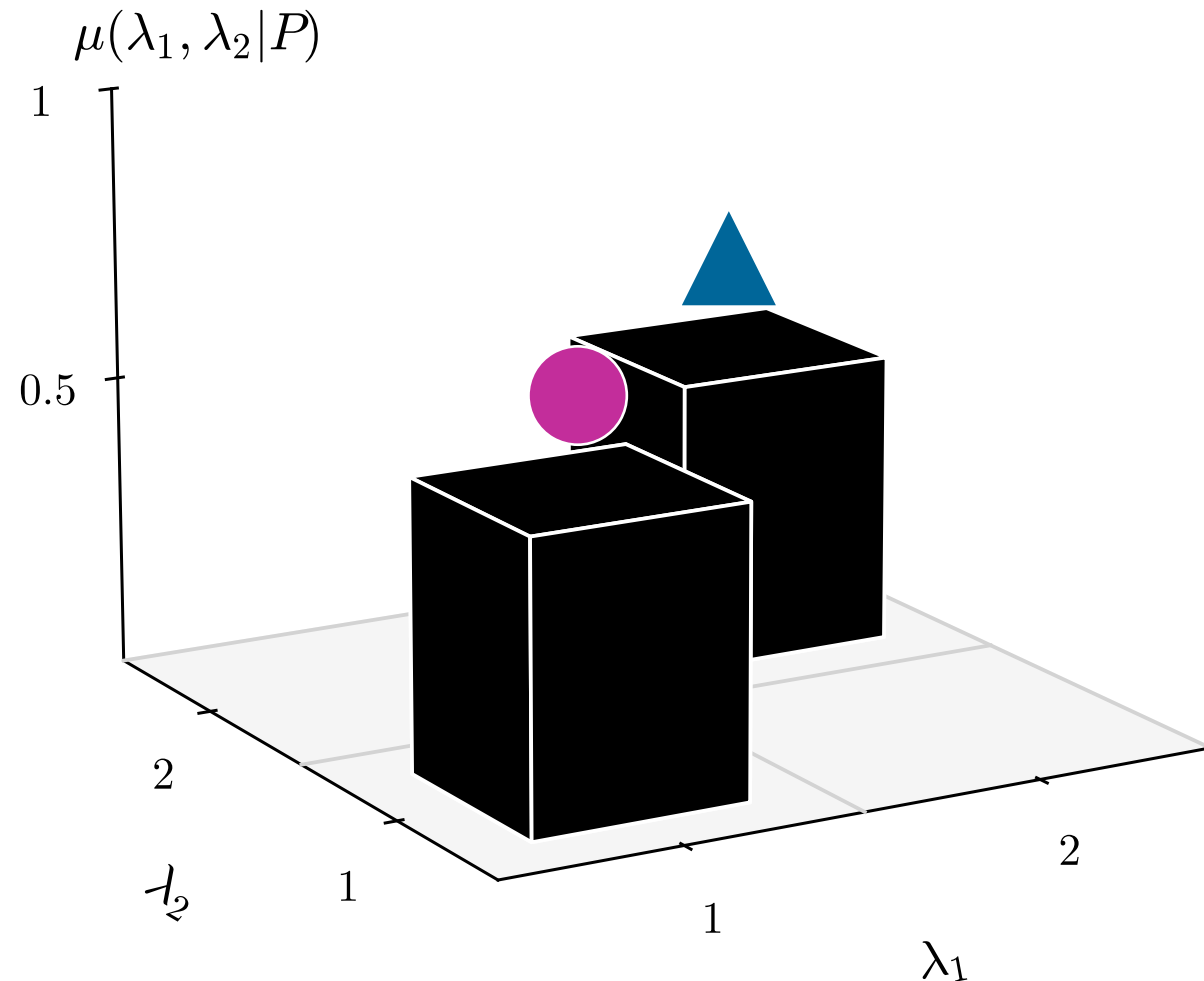
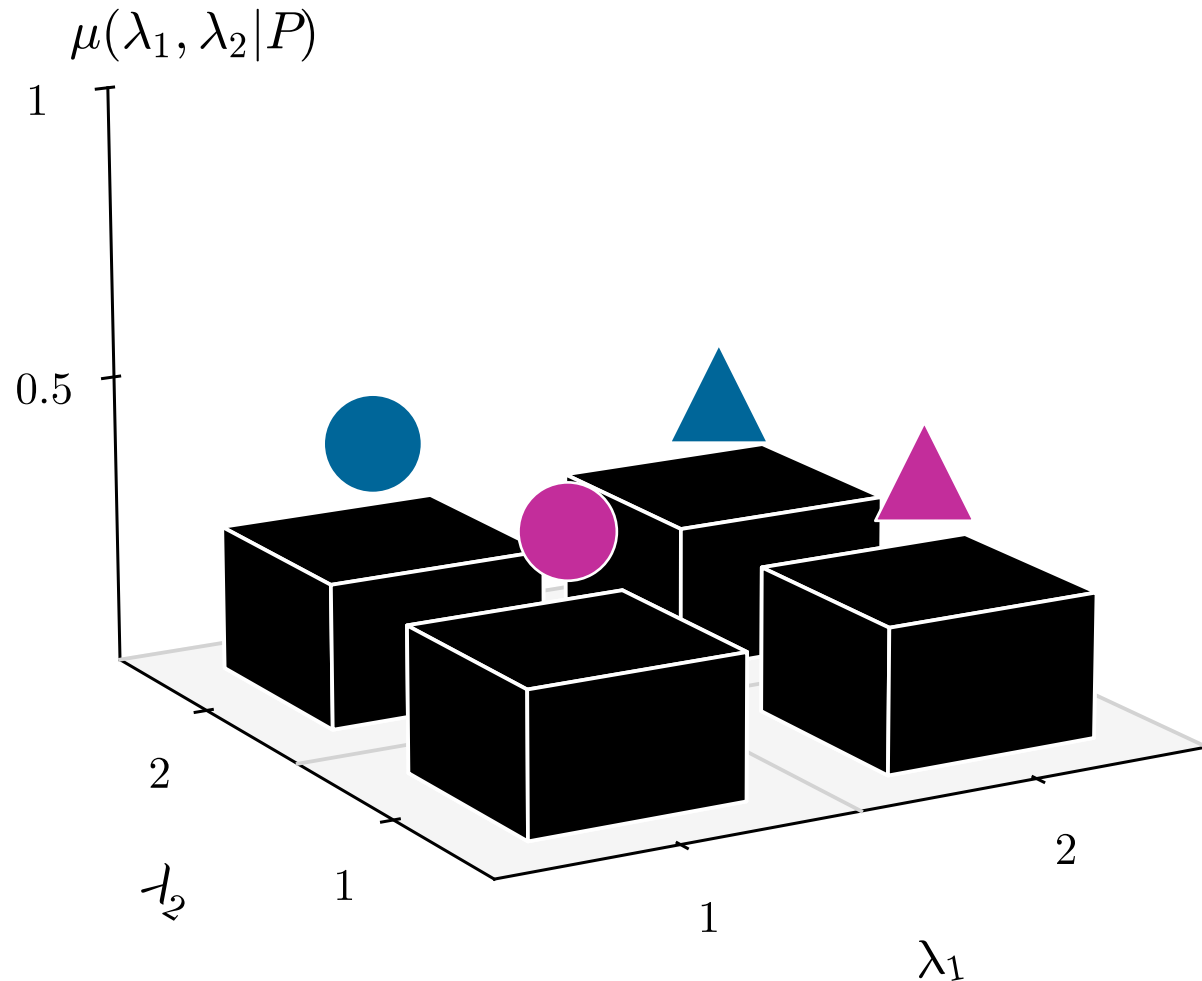
▲ 50% ● 50%

Outcome probabilities:

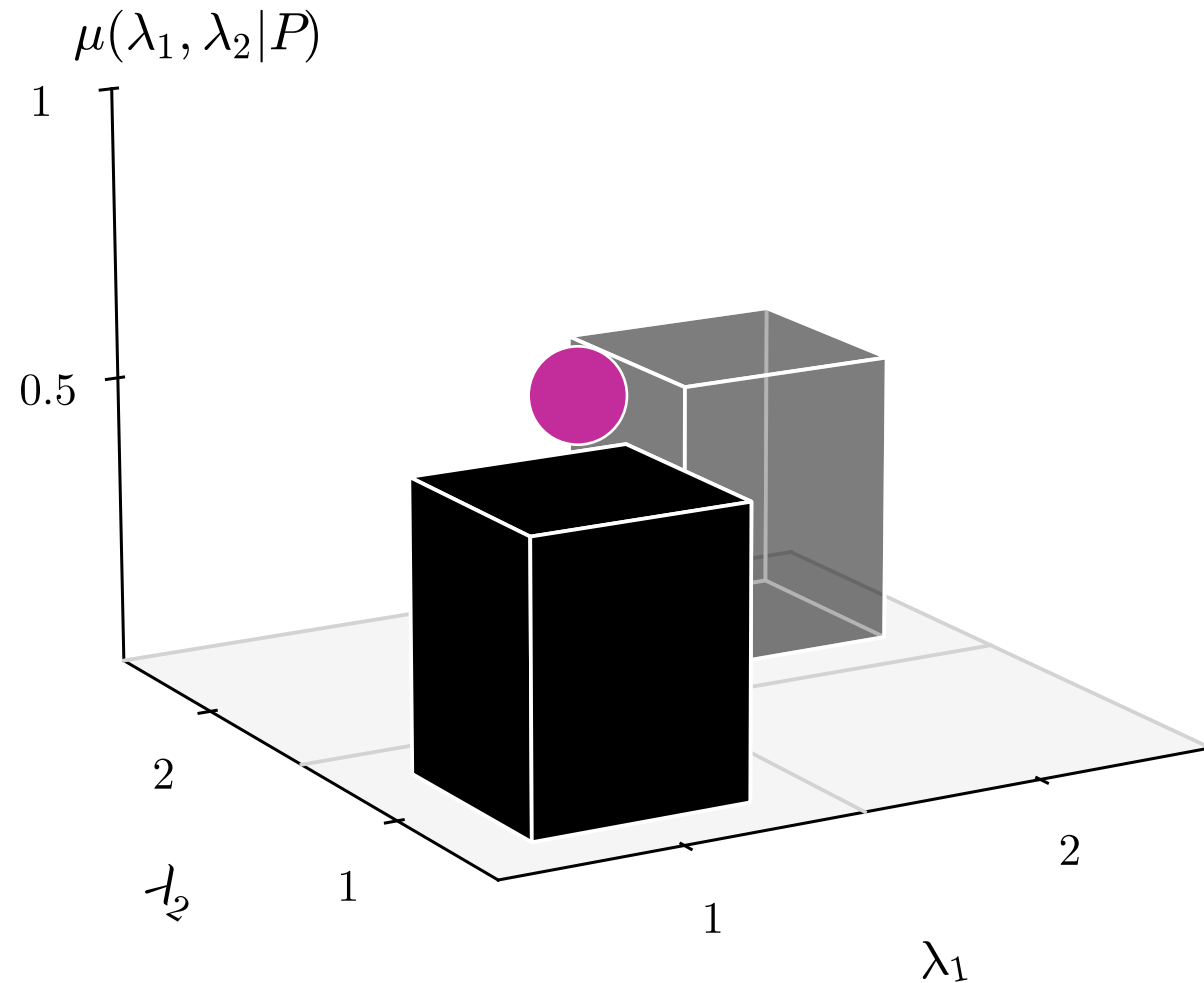
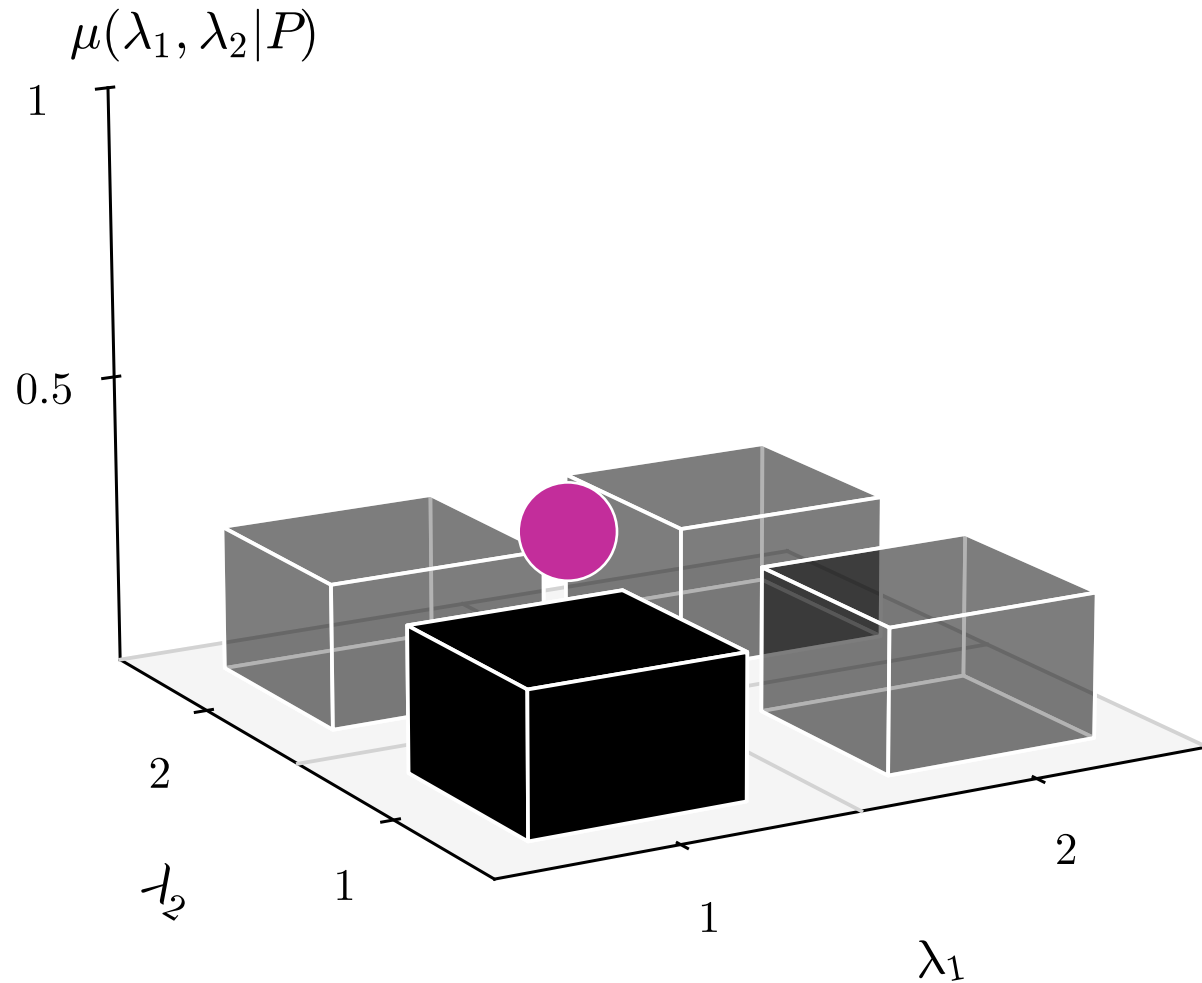
■ 50% ■ 50%



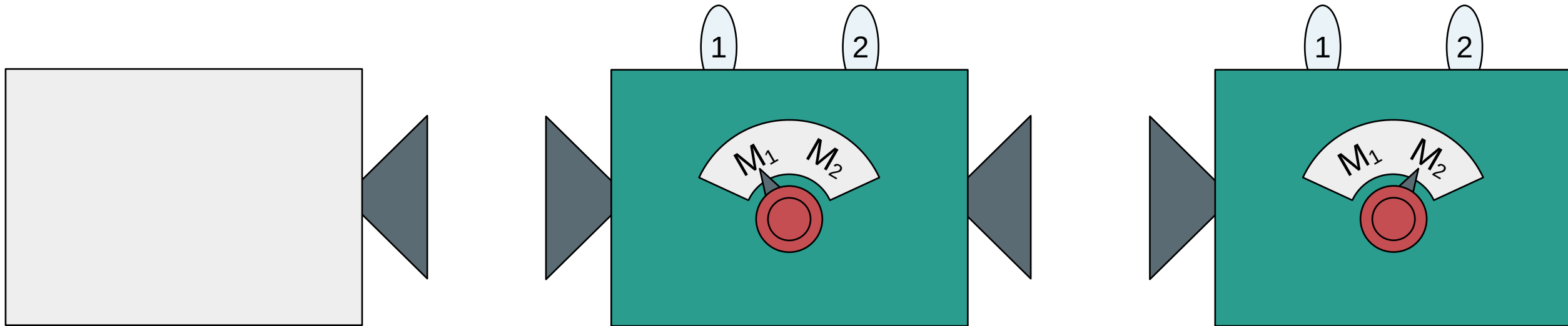
Probability distribution: $\mu(\lambda|P)$



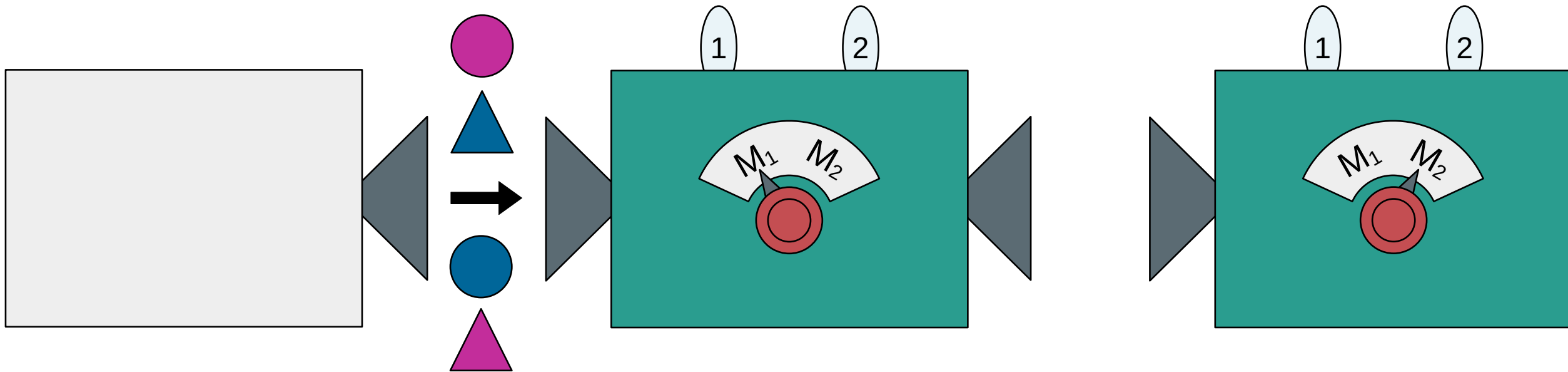
Probability distribution: $\mu(\lambda|P)$



Sequential (or simultaneous) measurement

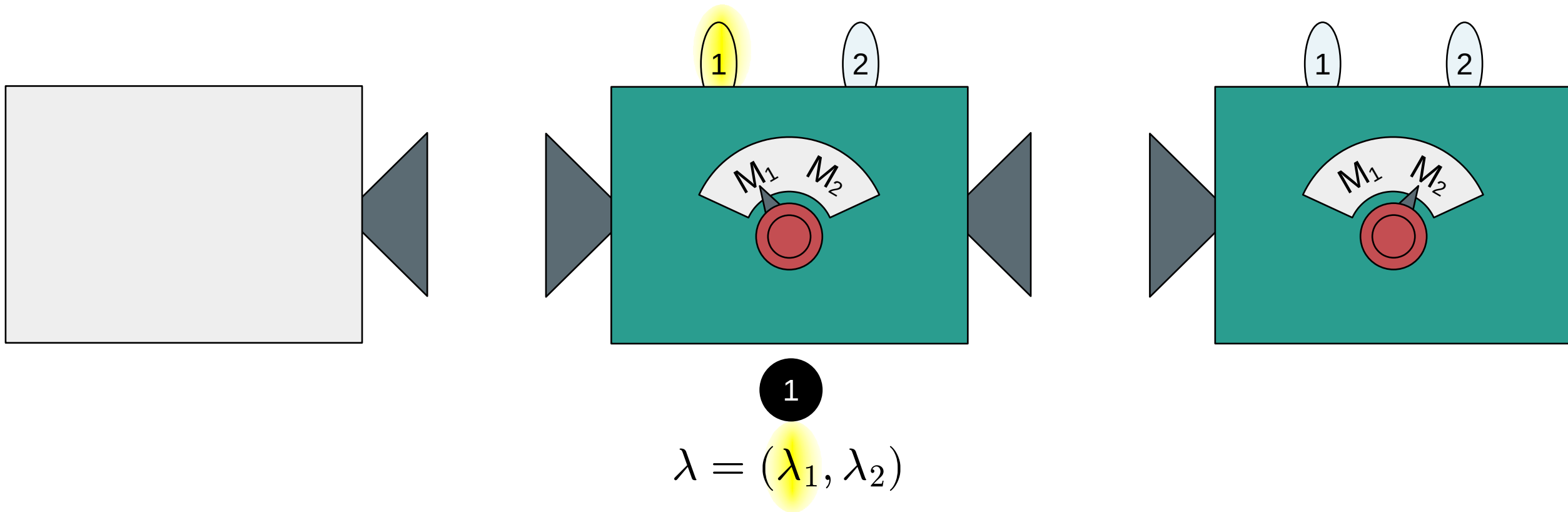


Sequential (or simultaneous) measurement

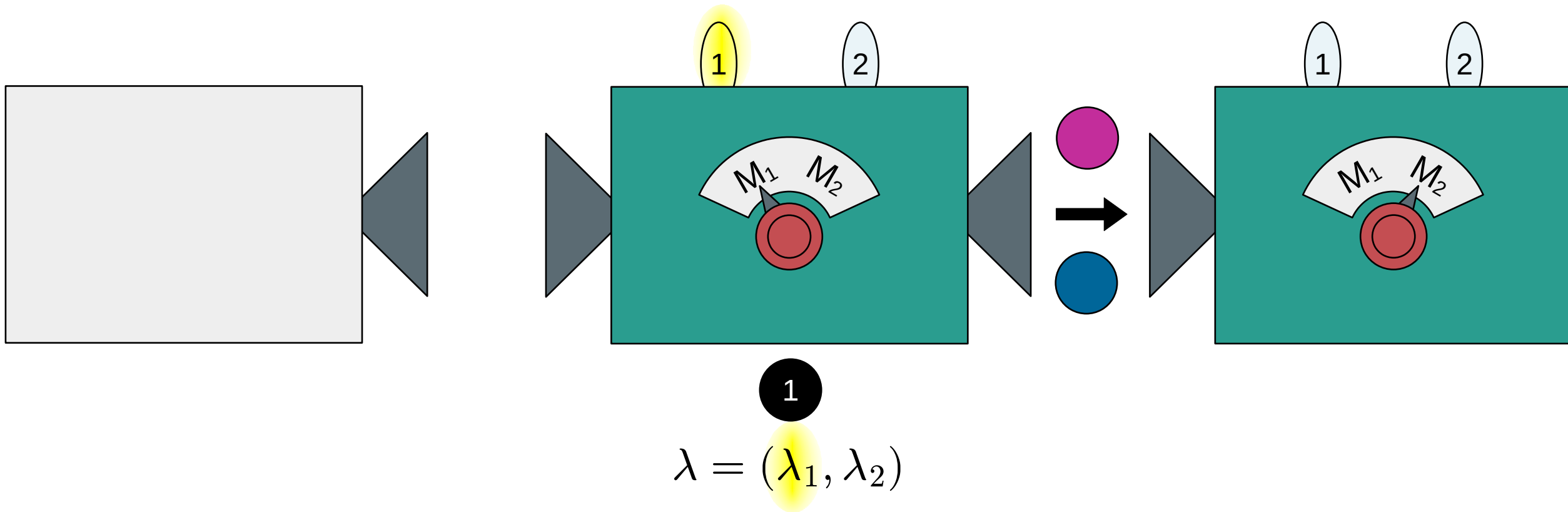


$$\lambda = (\lambda_1, \lambda_2)$$

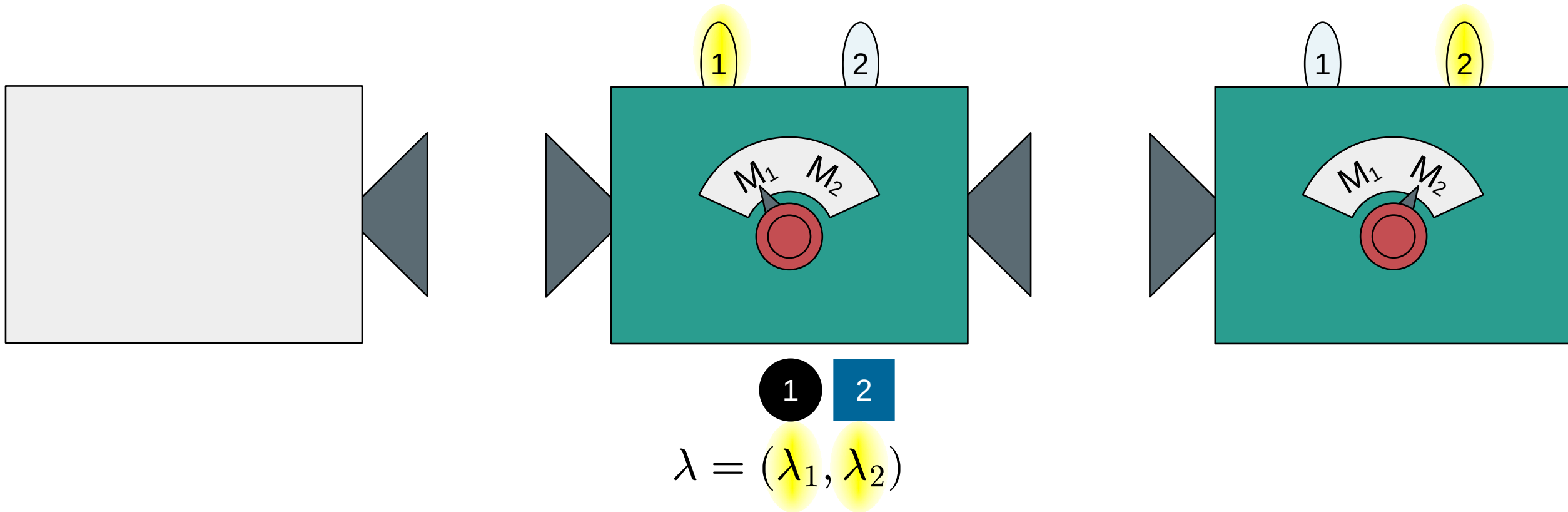
Sequential (or simultaneous) measurement



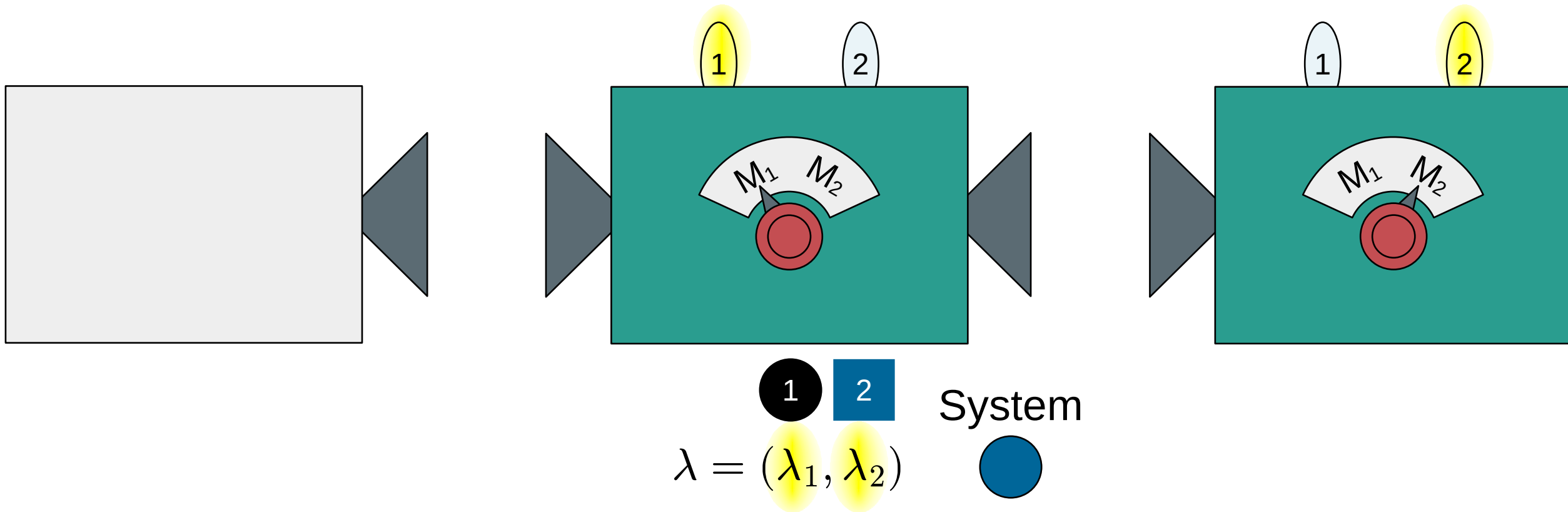
Sequential (or simultaneous) measurement



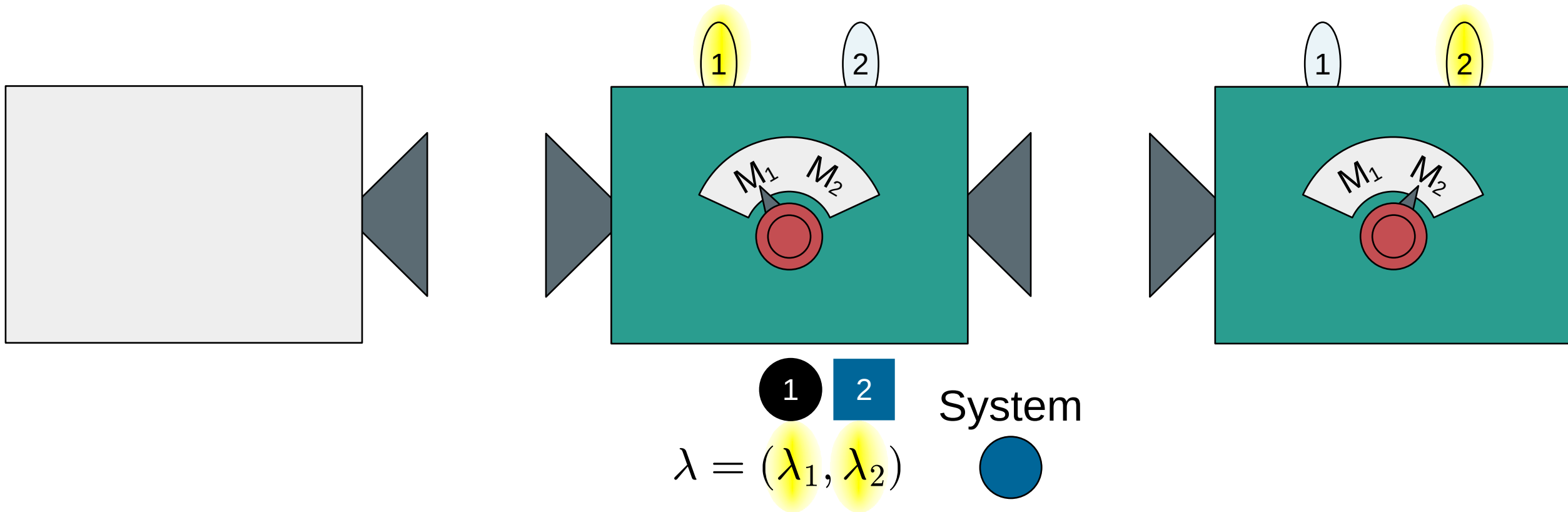
Sequential (or simultaneous) measurement



Sequential (or simultaneous) measurement



Sequential (or simultaneous) **non-disturbing** measurement



Outcome probabilities for non-disturbing measurements:

$$\text{Prob}(l, m, \dots | P, M_i, M_j, \dots) = \sum_{\lambda} \mu(\lambda | P) \xi(l | M_i, \lambda) \xi(m | M_j, \lambda) \dots$$

Outcome probabilities for non-disturbing measurements:

$$\text{Prob}(\underbrace{l, m, \dots}_k | P, \underbrace{M_i, M_j, \dots}_M) = \sum_{\lambda} \mu(\lambda | P) \underbrace{\xi(l | M_i, \lambda) \xi(m | M_j, \lambda) \dots}_{\xi(k | M, \lambda)}$$

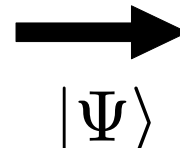
Outcome probabilities for non-disturbing measurements:

$$\text{Prob}(k|P, M) = \sum_{\lambda} \mu(\lambda|P) \xi(k|M, \lambda)$$

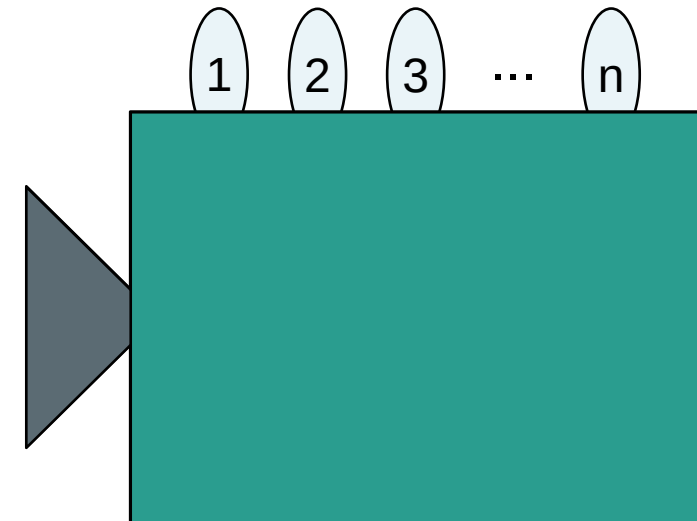
Preparation (P)



System



Measurement (M_j)



Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \dots)$

Preparation (P)

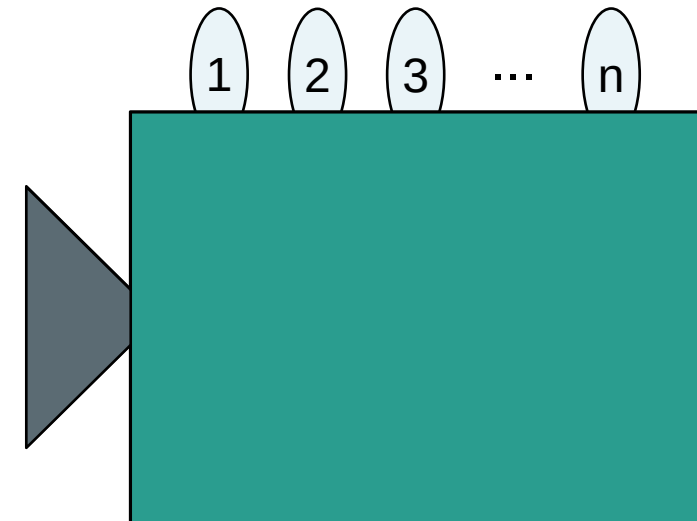


System



$|\Psi\rangle$

Measurement (M_j)



Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \dots)$

Superposition: $|\psi_j\rangle = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle + \dots$

Pure state: $|\Psi\rangle$

Preparation (P)



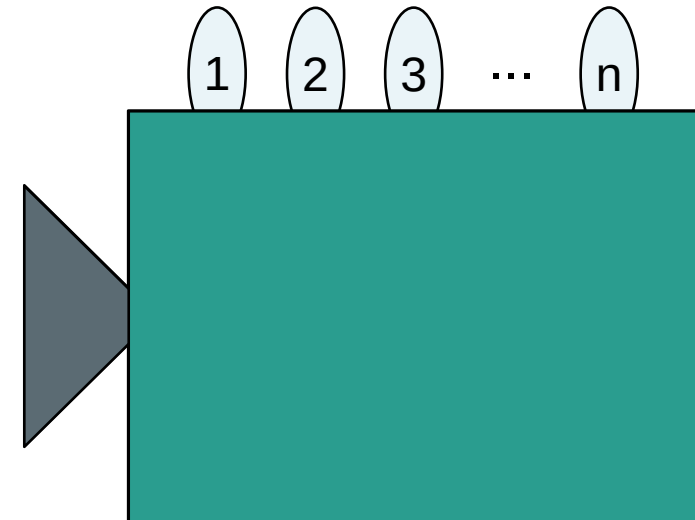
System



$|\Psi\rangle$

100%

Measurement (M_j)



Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)



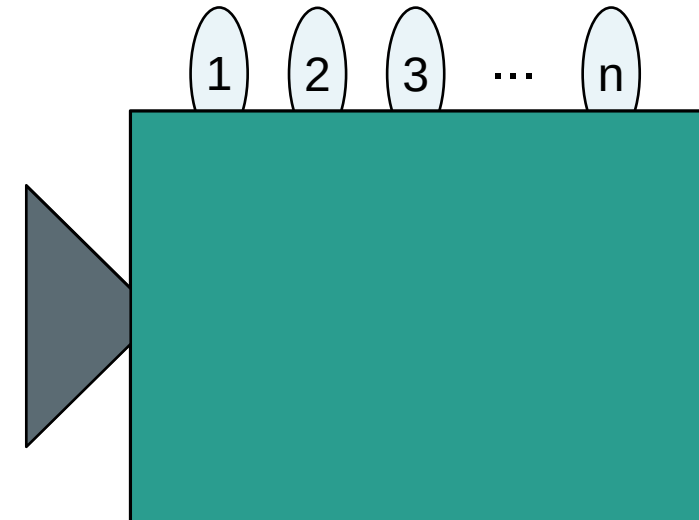
System



$|\Psi\rangle$

100%

Measurement (M_j)



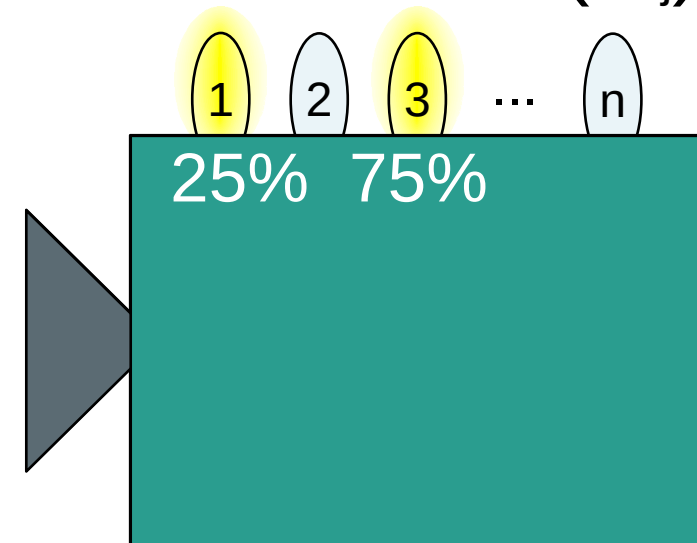
Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)



Measurement (M_j)



Outcome probabilities: $\text{Prob}(k|P, M) = \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle \leq 1$

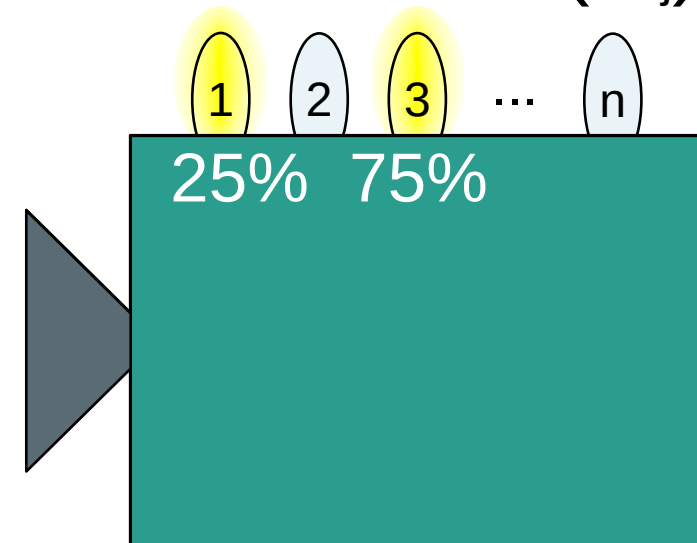
Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)



Measurement (M_j)



Is quantum mechanics secretly classical?

Outcome probabilities: $\text{Prob}(k|P, M) = \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle \leq 1$

Is quantum mechanics classical?

Pure state: $|\Psi\rangle$

Preparation (P)



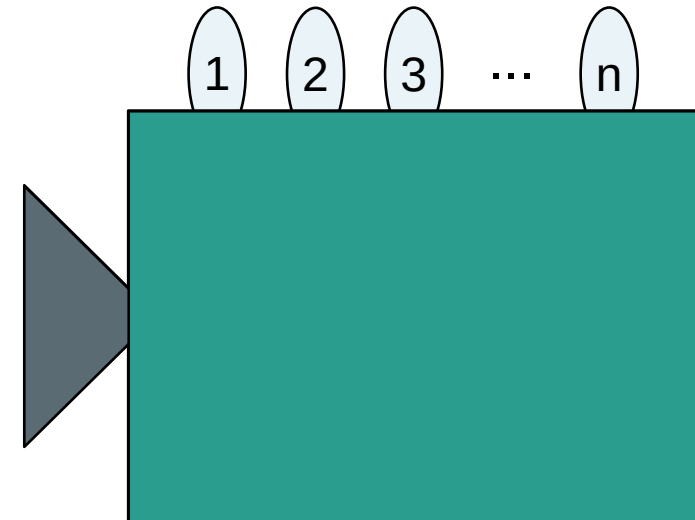
System



$|\Psi\rangle$

100%

Measurement (M_j)



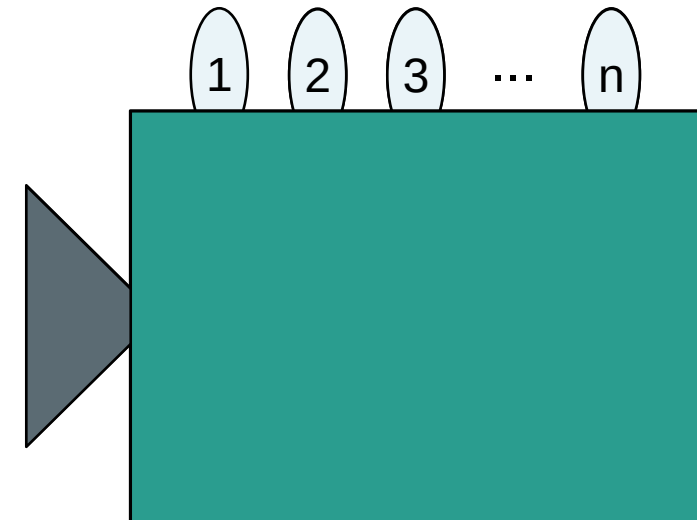
Probability distribution: $\mu(\lambda|P_i)$

Preparation (P)

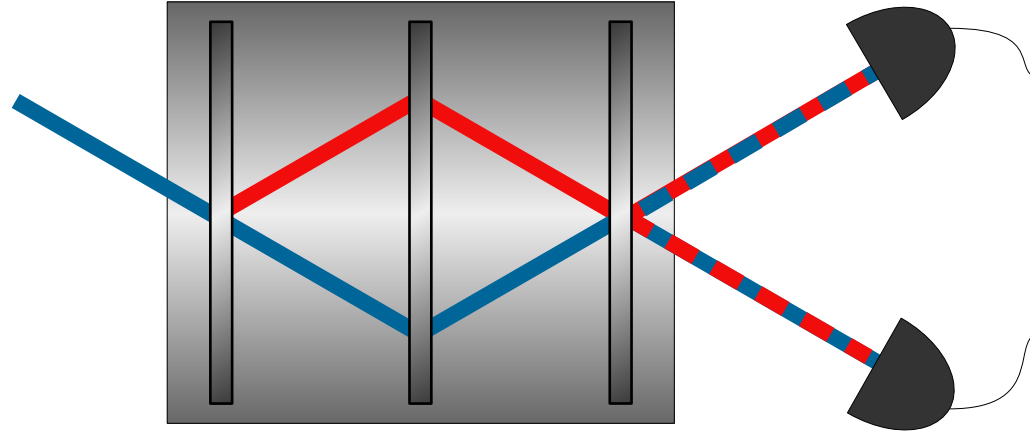


System
 λ' 75% λ'' 25%

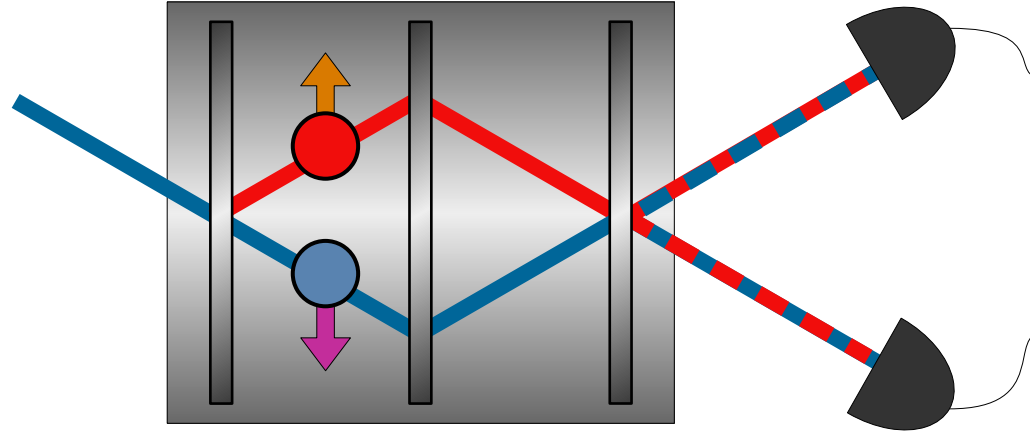
Measurement (M_j)



Neutron
Interferometer:



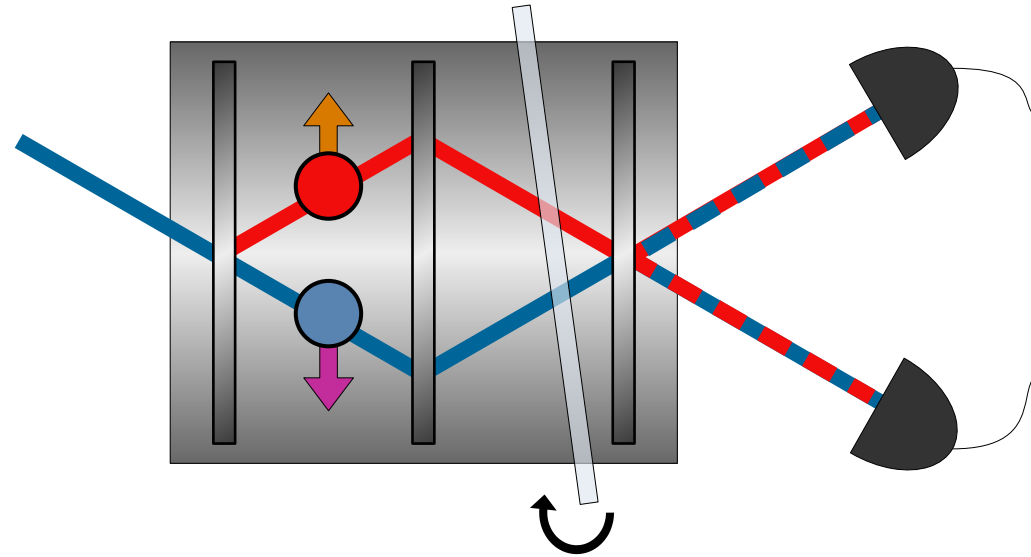
Neutron
Interferometer:



$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

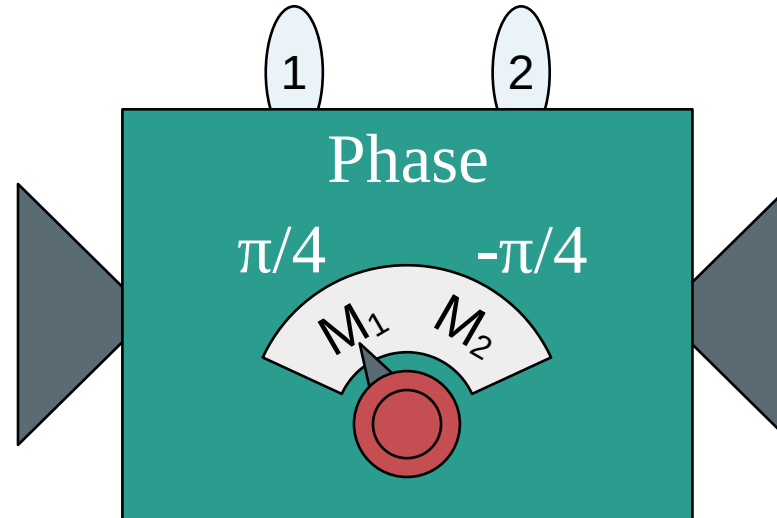
Preparation (P)

Neutron Interferometer:



$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

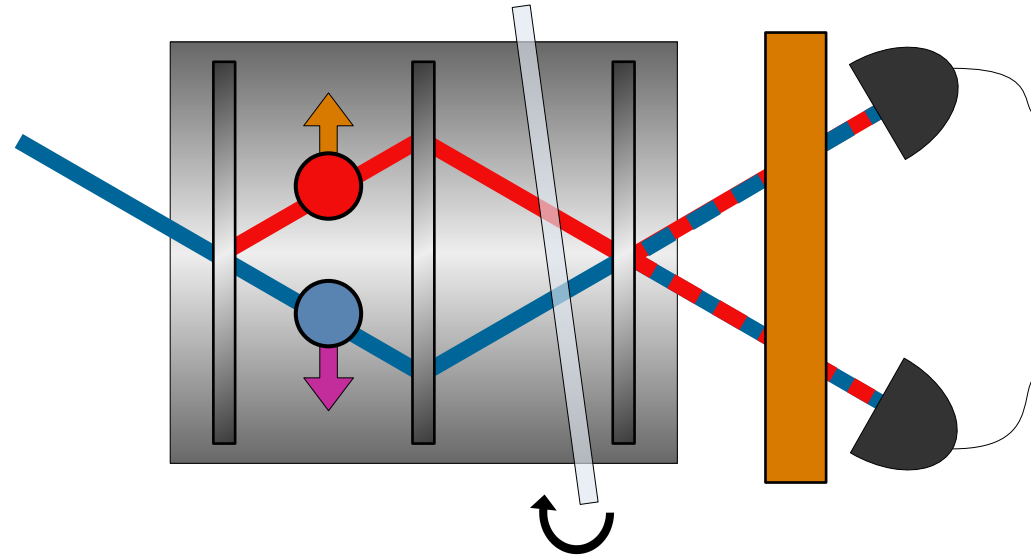
Preparation (P)



Path (M_1, M_2)

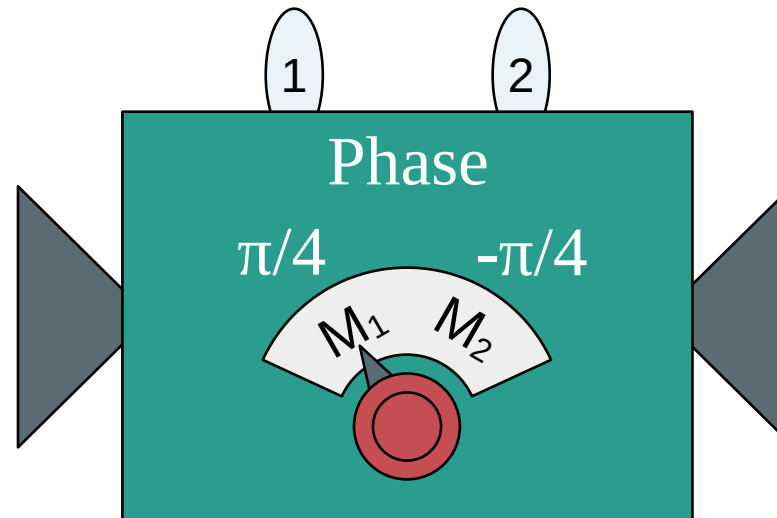
Neutron interferometry test of nonclassicality

Neutron
Interferometer:

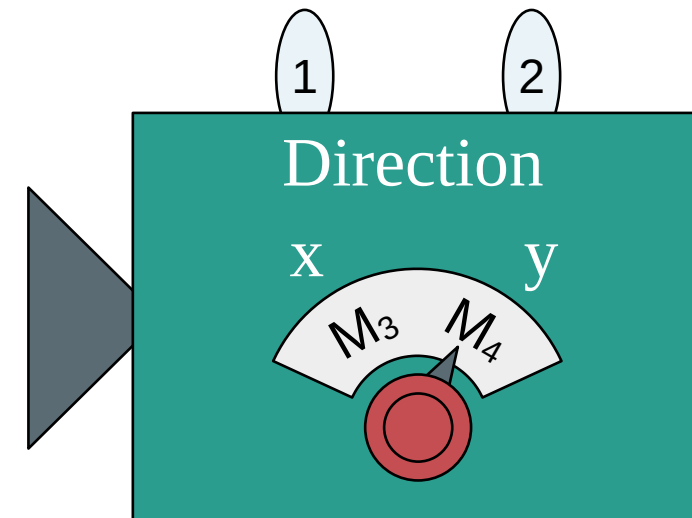


$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)



Path (M₁, M₂)



Spin (M₃, M₄)₅₈

Neutron interferometry test of nonclassicality

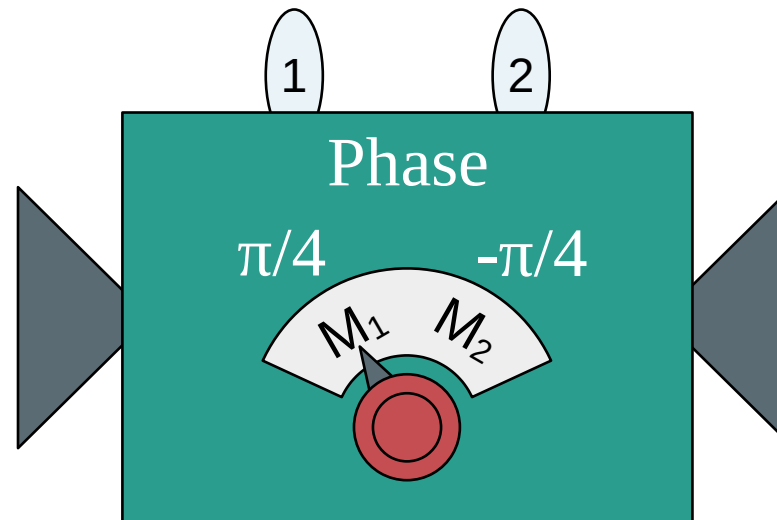
Measurements:

M_1 M_2

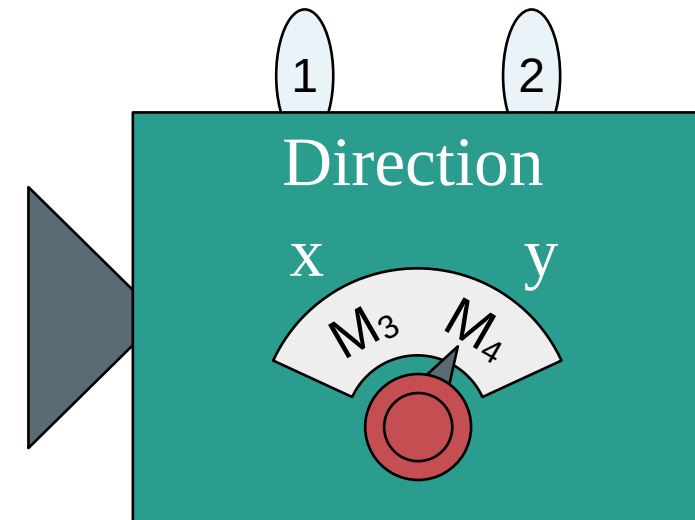
M_3 M_4

$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)



Path (M_1, M_2)



Spin (M_3, M_4)

Neutron interferometry test of nonclassicality

Measurements:

$M_1 \longleftrightarrow M_2$

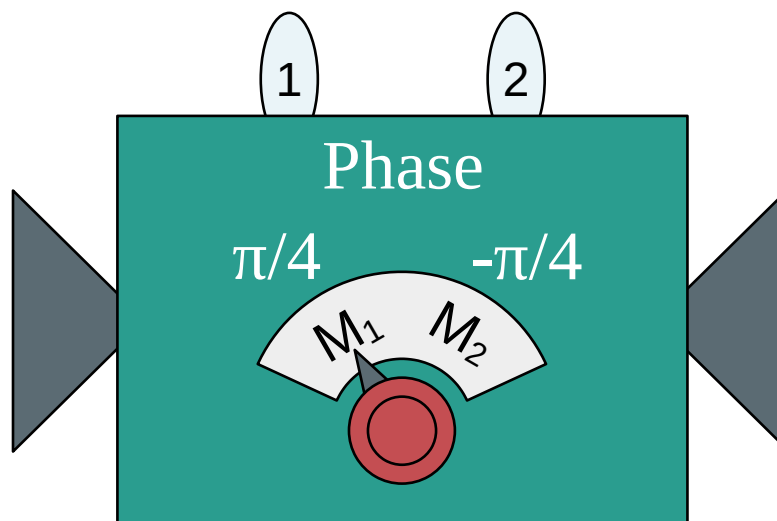
Disturbing
 $\longleftrightarrow$

$$[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}] \neq 0$$

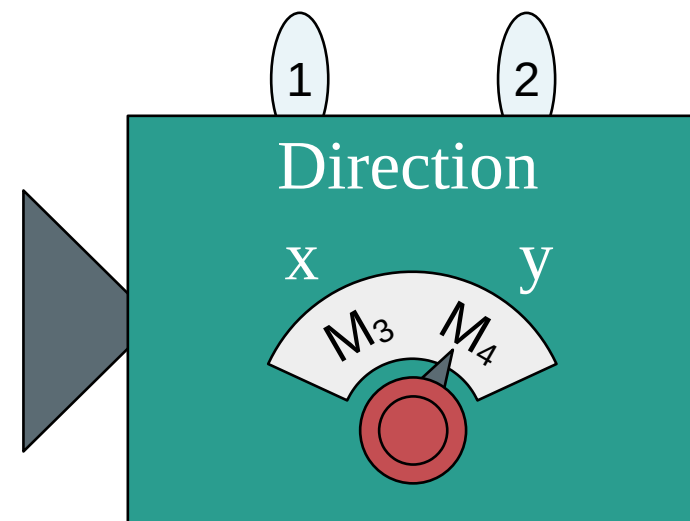
$M_3 \longleftrightarrow M_4$

$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)



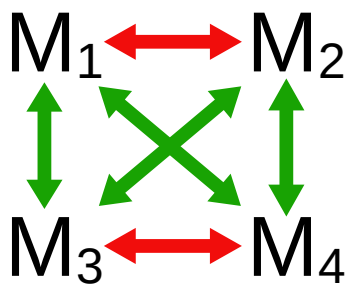
Path (M_1, M_2)



Spin (M_3, M_4)

Neutron interferometry test of nonclassicality

Measurements:

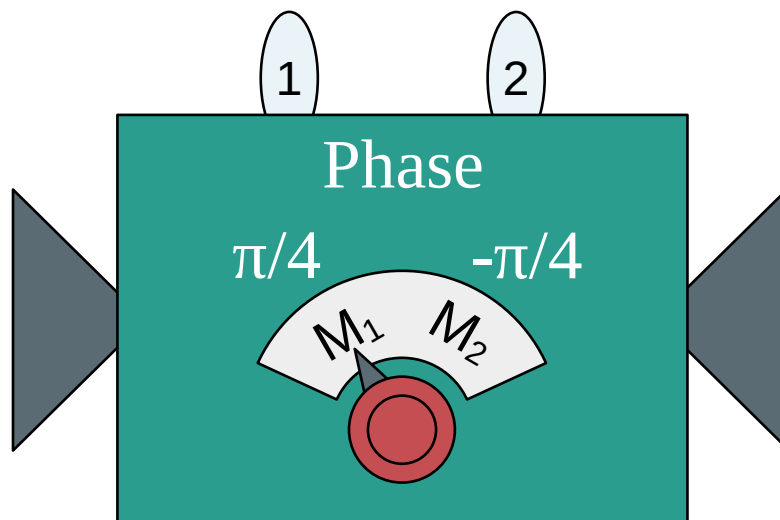


Disturbing
Non-disturbing

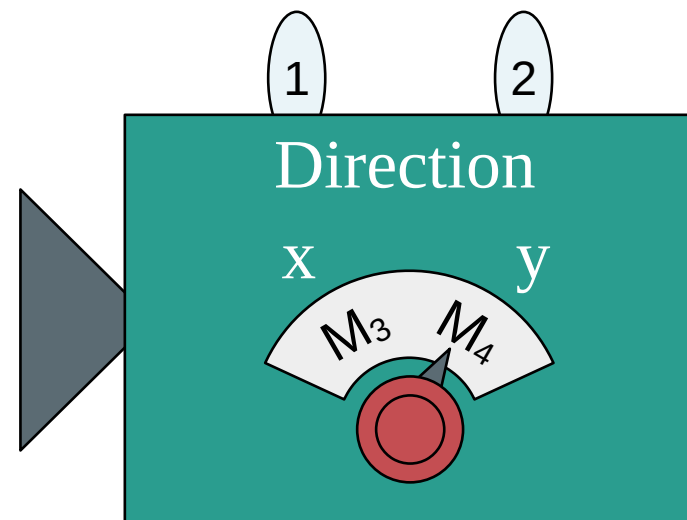
$$\begin{aligned} [\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}] &\neq 0 \\ [\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}] &= 0 \end{aligned}$$

$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)



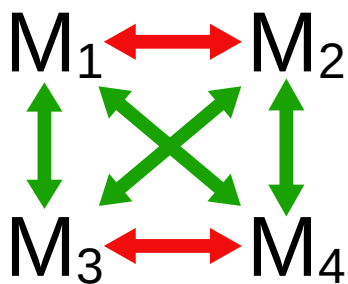
Path (M_1, M_2)



Spin (M_3, M_4)

Neutron interferometry test of nonclassicality

Measurements:

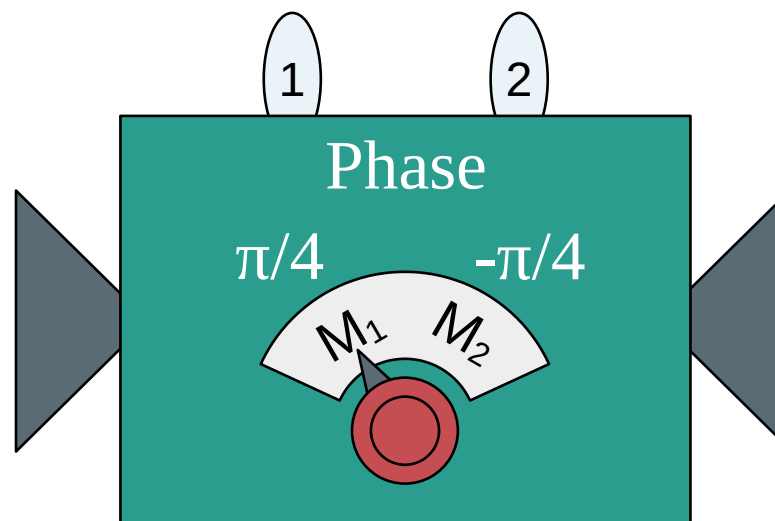


Disturbing
Non-disturbing

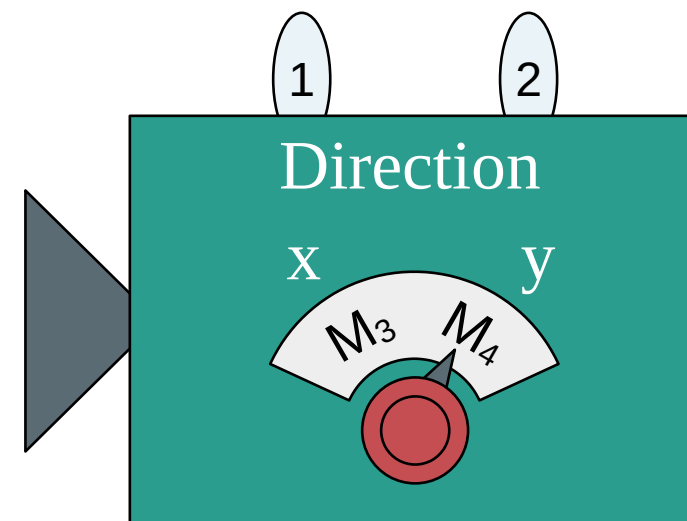
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$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

Preparation (P)



Path (M_1, M_2)



Spin (M_3, M_4)

Noncontextuality

Outcome probabilities:

$$\text{Prob}(k, l|P, M_i, M_j) = \langle \Psi | \hat{\Pi}_{k|M_i} \hat{\Pi}_{l|M_j} | \Psi \rangle$$

$$\stackrel{?}{=} \sum_{\lambda} \mu(\lambda|P) \xi(k|M_i, \lambda) \xi(l|M_j, \lambda)$$

Outcomes correlations:

$$E(M_i, M_j) = \text{Prob}(k, k|P, M_i, M_j) - \text{Prob}(k, l \neq k|P, M_i, M_j)$$

Clauser–Horne–Shimony–Holt (CHSH) inequality:

$$E(M_1, M_3) - E(M_1, M_4) + E(M_2, M_3) + E(M_2, M_4) \leq 2$$

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Experimental results:

$$2.051 \pm 0.0019$$

Hasegawa, Yuji, et al. "Violation of a bell-like inequality in single-neutron interferometry." Nature 425, 45–48 (2003).

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Hasegawa, Yuji, et al. "Violation of a bell-like inequality in single-neutron interferometry." *Nature* 425, 45–48 (2003).

$$2.365 \pm 0.0013$$

Geppert, Hermann, et al. "Improvement of the polarized neutron interferometer setup demonstrating violation of a Bell-like inequality." *Nuclear Instruments and Methods in Physics Research Section A* 763 (2014): 417-423.

Is quantum mechanics secretly classical?

...NO!

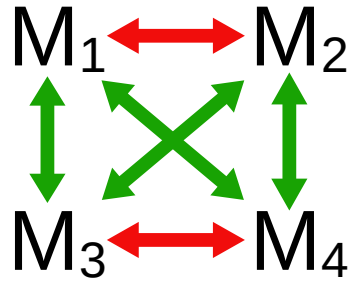
Is quantum mechanics secretly classical?

...NO!*

*according to our definition of non-disturbance (commutativity)

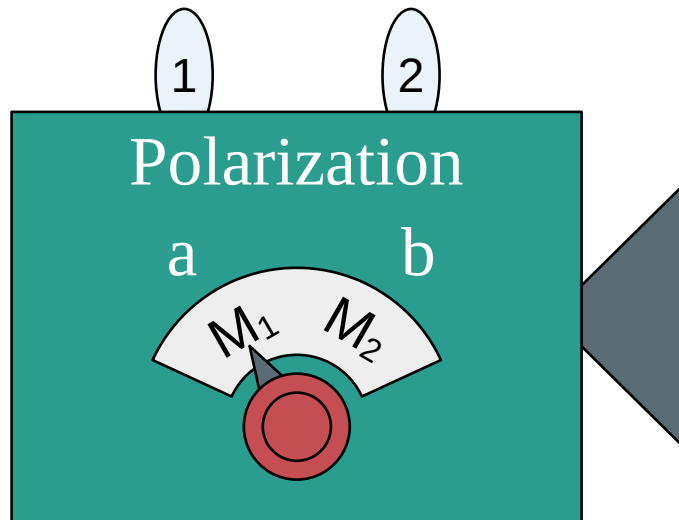
Bell test of nonclassicality

Measurements:

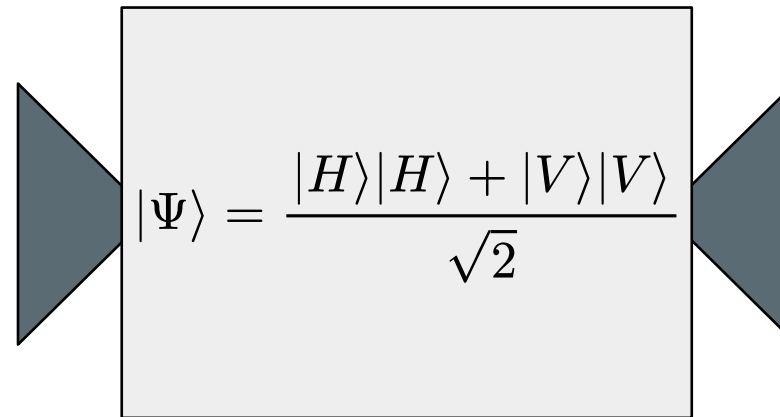


Disturbing
 $\longleftrightarrow$
 Non-disturbing
 $\longleftrightarrow$

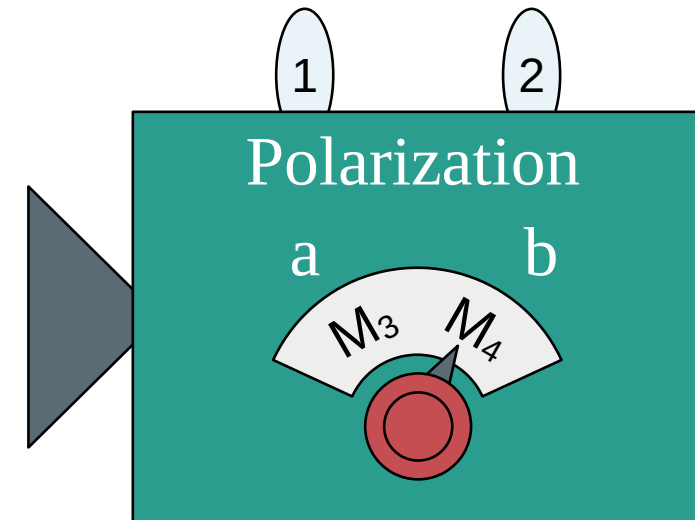
$$\begin{aligned} [\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}] &\neq 0 \\ [\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}] &= 0 \end{aligned}$$



Particle 1 (M_1, M_2)



Preparation (P)



Particle 2 (M_3, M_4)

Is quantum mechanics secretly classical?

Still...NO!*

*independently of the quantum formulation

Quasiprobability representations and weak values

Outcome probabilities:

$$\text{Prob}(k|P, M) = \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle \stackrel{?}{=} \sum_{\lambda} \mu(\lambda|P) \xi(k|M, \lambda)$$

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

Kirkwood-Dirac quasiprobability representation:

$$\text{Prob}(k|P, M) = \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle = \sum_{a_i, b_j} \langle \Psi | b_j \rangle \langle b_j | a_i \rangle \langle a_i | \Psi \rangle \frac{\langle b_j | \hat{\Pi}_{k|M} | a_i \rangle}{\langle b_j | a_i \rangle}$$

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Kirkwood-Dirac quasiprobability representation:

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Anomalous features:

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda, \hat{\Pi}_{k|M}) < 0$$

$$\tilde{\xi}(k|\lambda, \hat{\Pi}_{k|M}) > 1$$

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda, \hat{\Pi}_{k|M}) \in \mathbb{C}$$

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

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$$\text{Prob}(k|P, M) = \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle = \sum_{a_i, b_j} \underbrace{\langle \Psi | b_j \rangle \langle b_j | a_i \rangle \langle a_i | \Psi \rangle}_{\text{Quasiprobability distribution}} \frac{\langle b_j | \hat{\Pi}_{k|M} | a_i \rangle}{\underbrace{\langle b_j | a_i \rangle}_{\text{Symbol}}}$$

Anomalous features:

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..and we can measure it!

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

Kirkwood-Dirac quasiprobability representation:

$$\begin{aligned}\text{Prob}(k|P, M) &= \langle \Psi | \hat{\Pi}_{k|M} | \Psi \rangle = \sum_{a_i, b_j} \langle \Psi | b_j \rangle \langle b_j | a_i \rangle \langle a_i | \Psi \rangle \frac{\langle b_j | \hat{\Pi}_{k|M} | a_i \rangle}{\langle b_j | a_i \rangle} \\ &= \sum_{a_i, b_j} |\langle b_j | \Psi \rangle|^2 \frac{\langle b_j | a_i \rangle \langle a_i | \Psi \rangle}{\langle b_j | \Psi \rangle} \frac{\langle b_j | \hat{\Pi}_{k|M} | a_i \rangle}{\langle b_j | a_i \rangle}\end{aligned}$$

Weak values!

..and we can measure them!

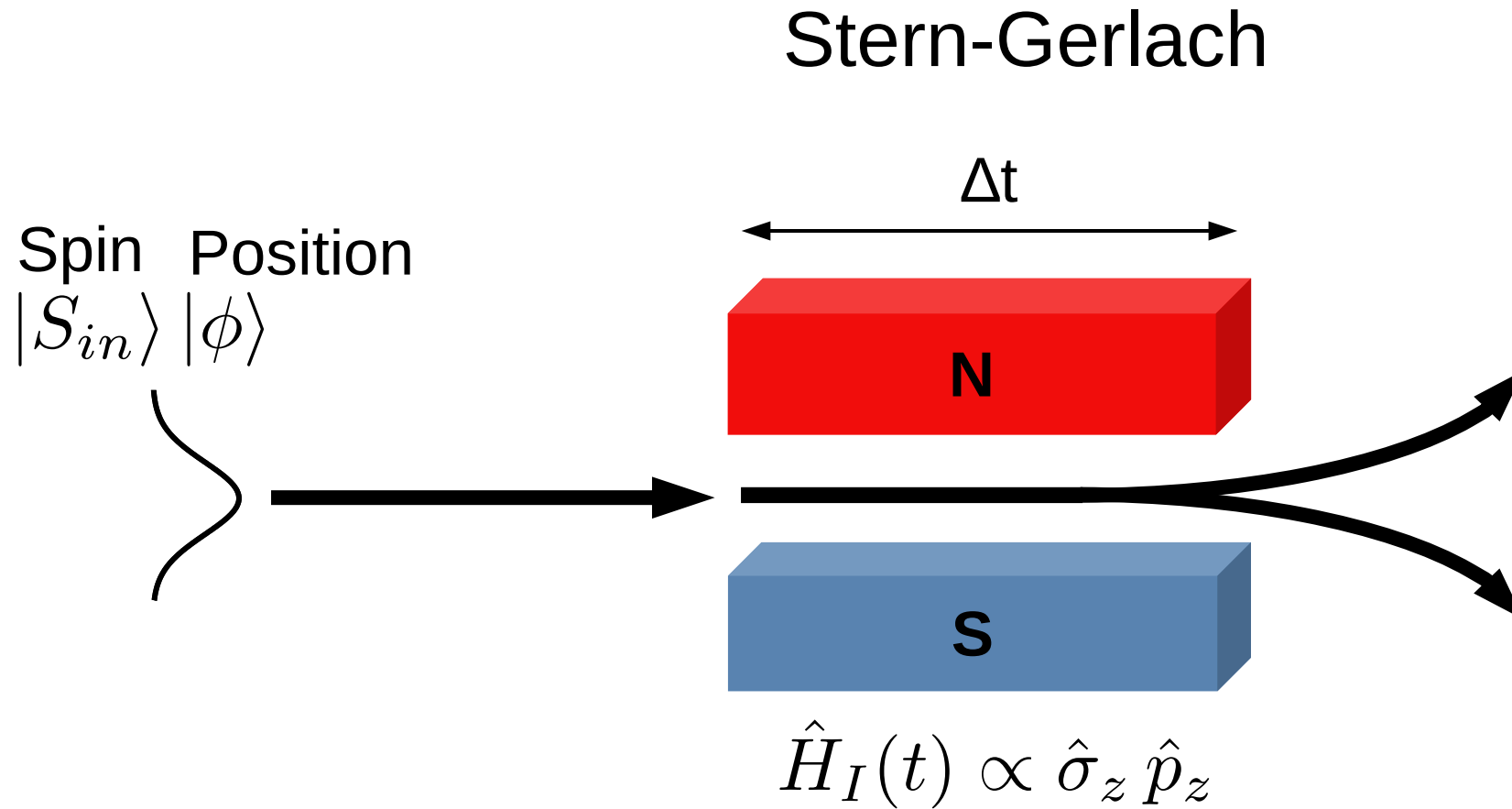
Stern-Gerlach

Spin Position

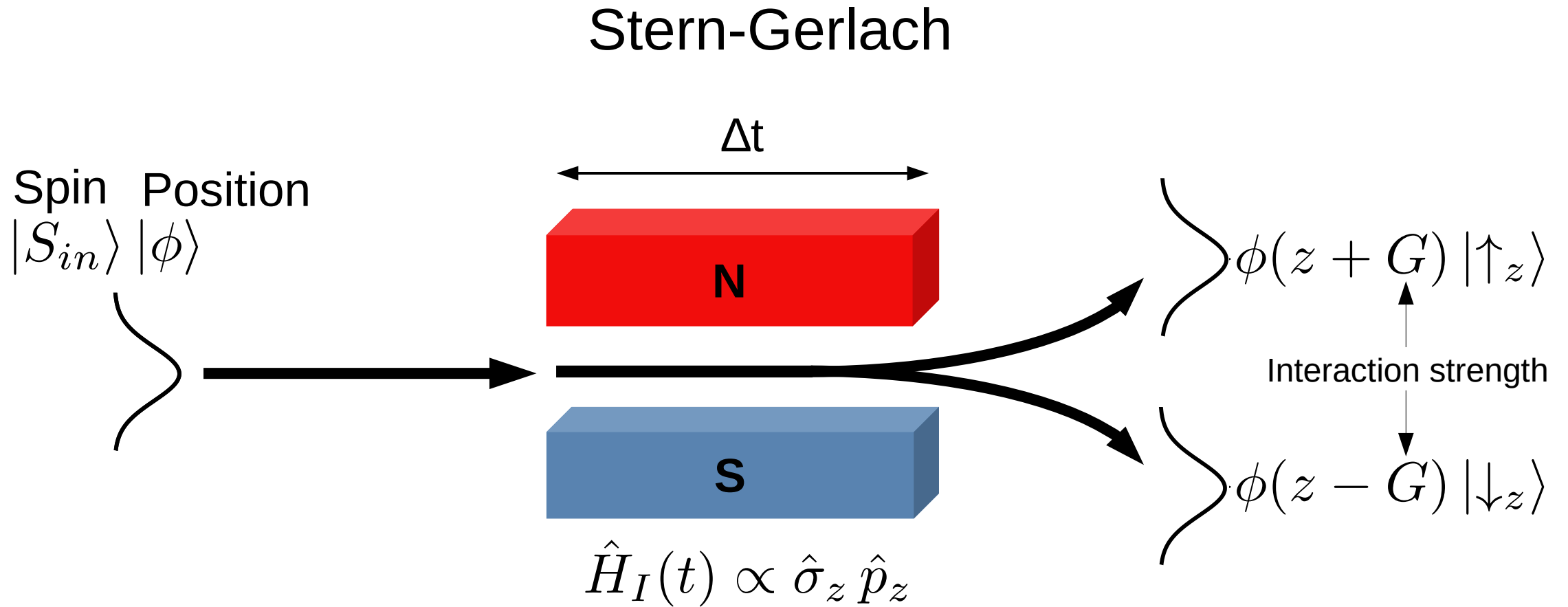
$|S_{in}\rangle |\phi\rangle$



A standard (Von Neumann) measurement

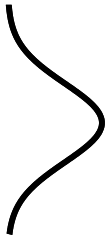


A standard (Von Neumann) measurement

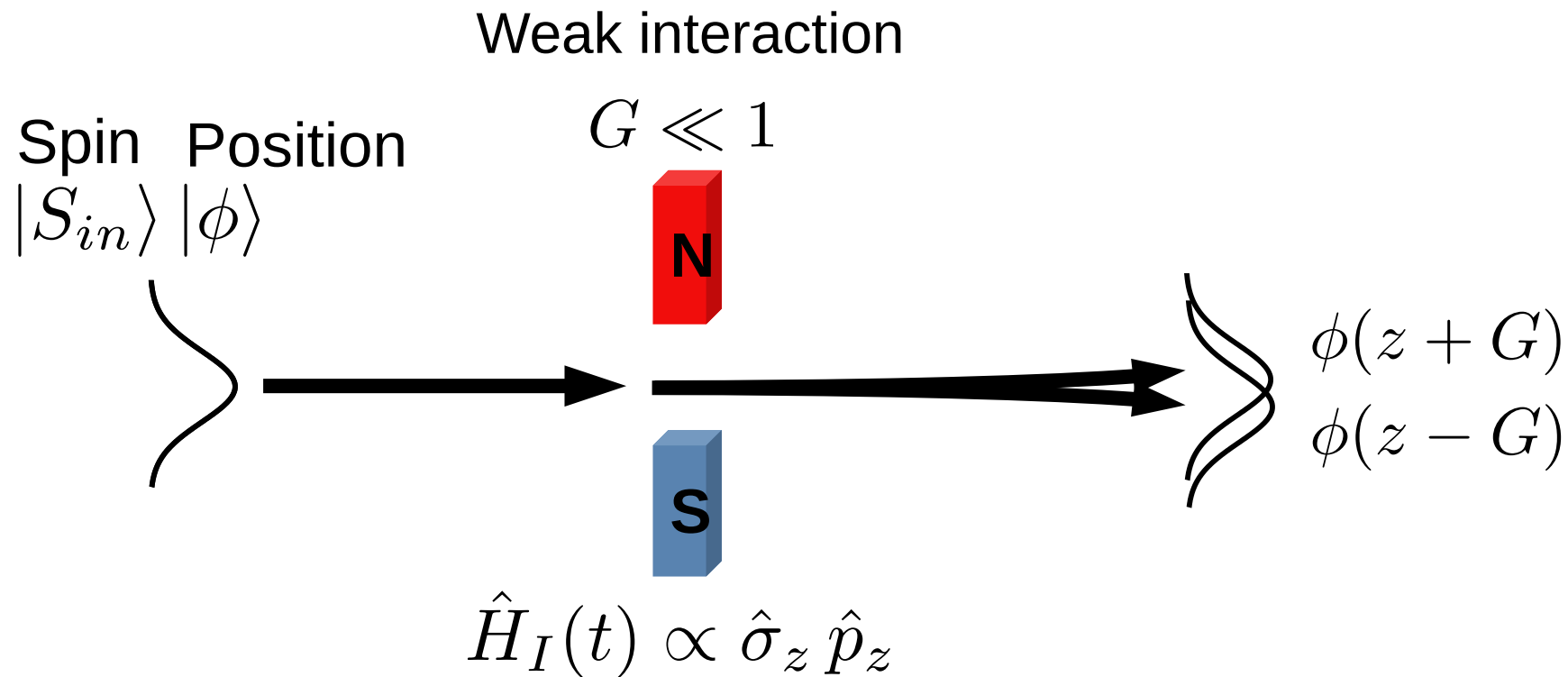


Spin Position

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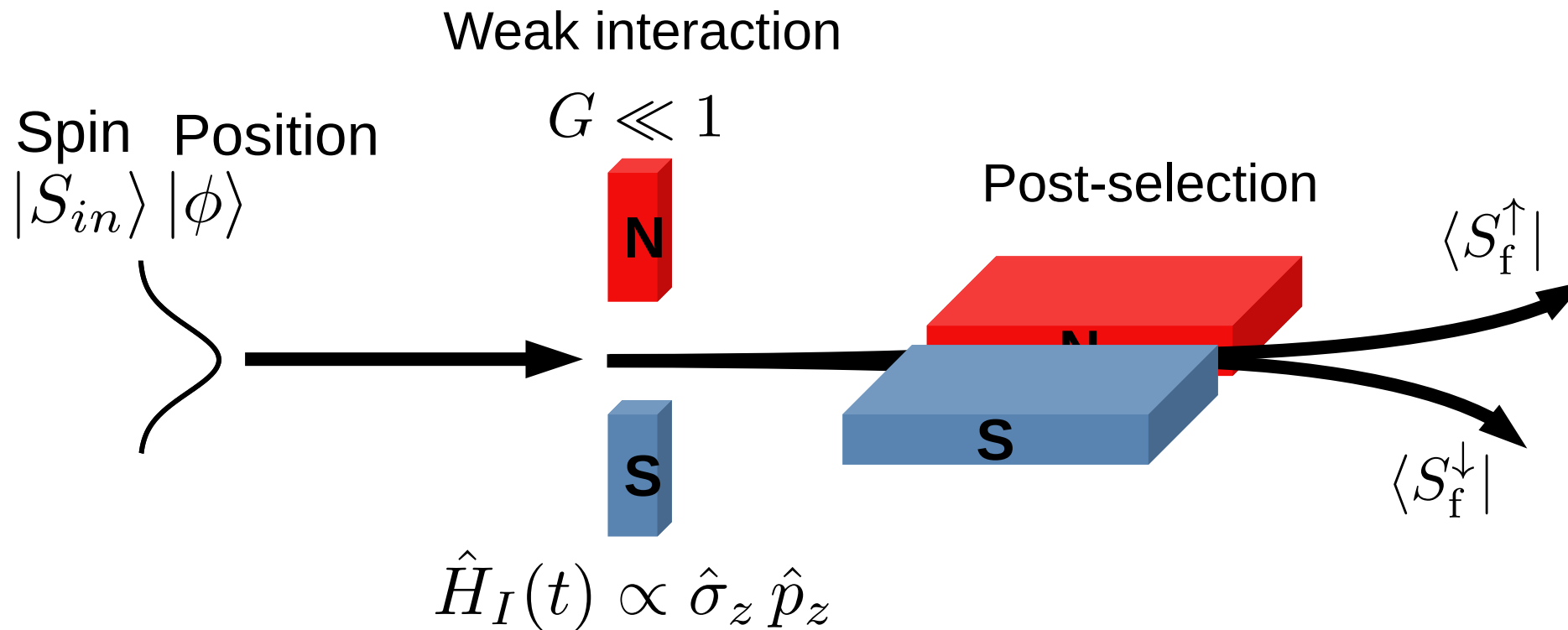


A weak measurement



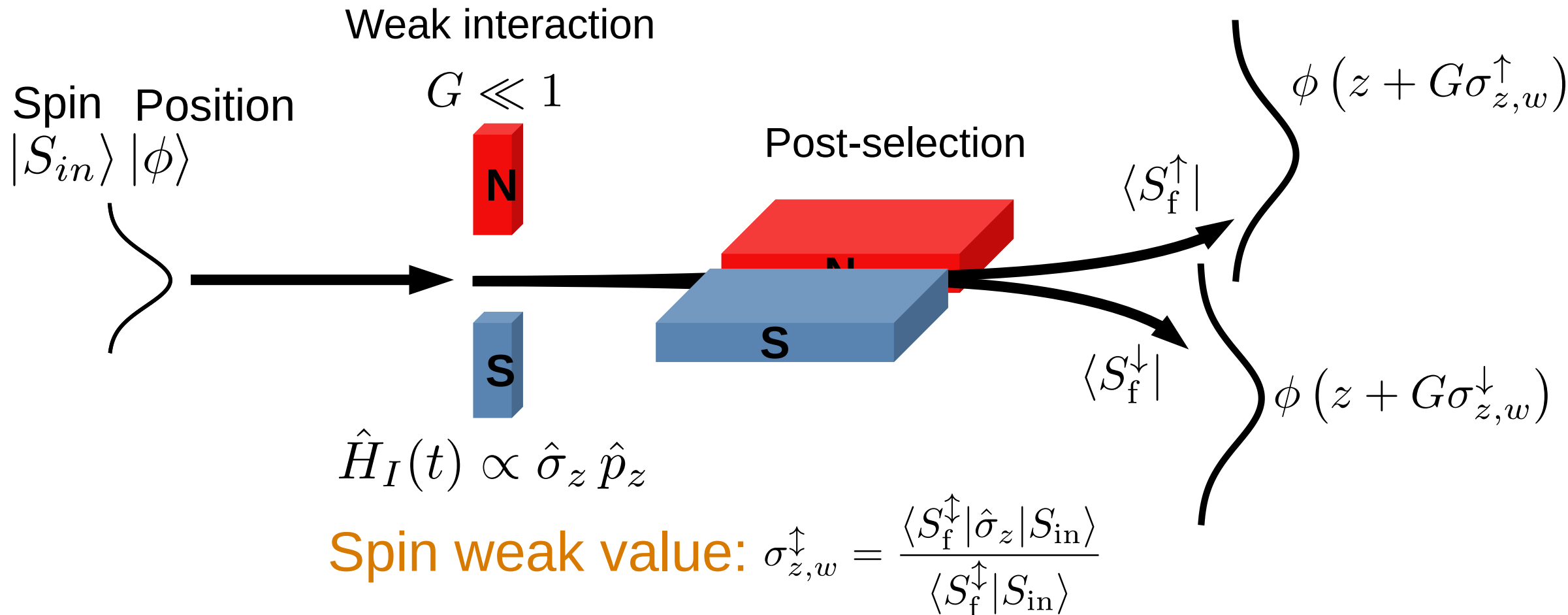
Aharonov, Yakir, et al. "How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100." *Physical review letters* 60.14 (1988): 1351.

A weak measurement



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Meter state and weak interaction:

- Lemmel, Hartmut, et al. "Quantifying the presence of a neutron in the paths of an interferometer." *Physical Review Research* 4.2 (2022): 023075.
- Danner, Armin, et al. "Simultaneous path weak-measurements in neutron interferometry." *Scientific Reports* 14.1 (2024): 25994.
- Dvorak, Andreas, et al. "Tight qubit uncertainty relations studied through weak values in neutron interferometry." *Physical Review Research* 7.4 (2025): 043334.

Meter state and no weak interaction:

- Denkmayr, Tobias, et al. "Weak values from strong interactions in neutron interferometry." *Physica B: Condensed Matter* 551 (2018): 339-346.

No meter state and weak interaction:

- Masiello, Ismaele V., et al. "Simultaneous determination of two path weak-values with time-dependent phase manipulation in neutron interferometry." *New Journal of Physics* 27.2 (2025): 023017.

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Meter states or weak interactions increase
the experimental complexity!

The manipulation of additional degrees of freedom increases the amount of required experimental resources (additional instrumentation and procedures), while weak interactions typically necessitate longer measurement times due to the limited information gained per detection event.

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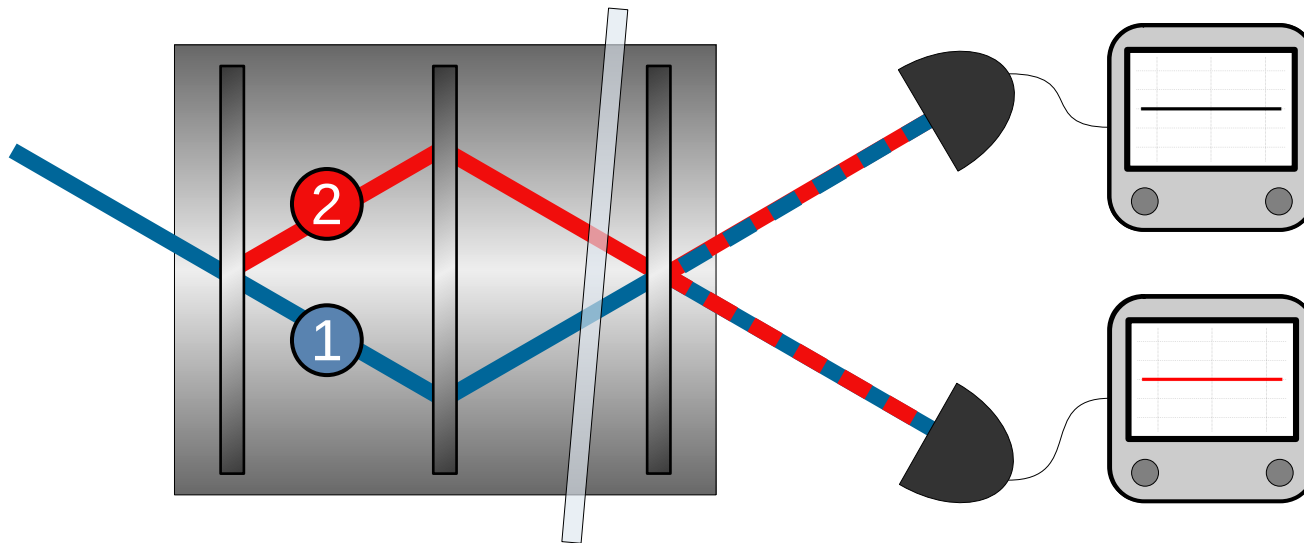
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No meter state and no weak interaction:

- *Masiello, Ismaele V., et al. "Anomalous weak values in a generalized Mach-Zehnder interferometer extracted directly from intensity measurements." in review at npj Quantum Information.*

Experimental results

Generalized neutron Mach-Zehnder interferometer:



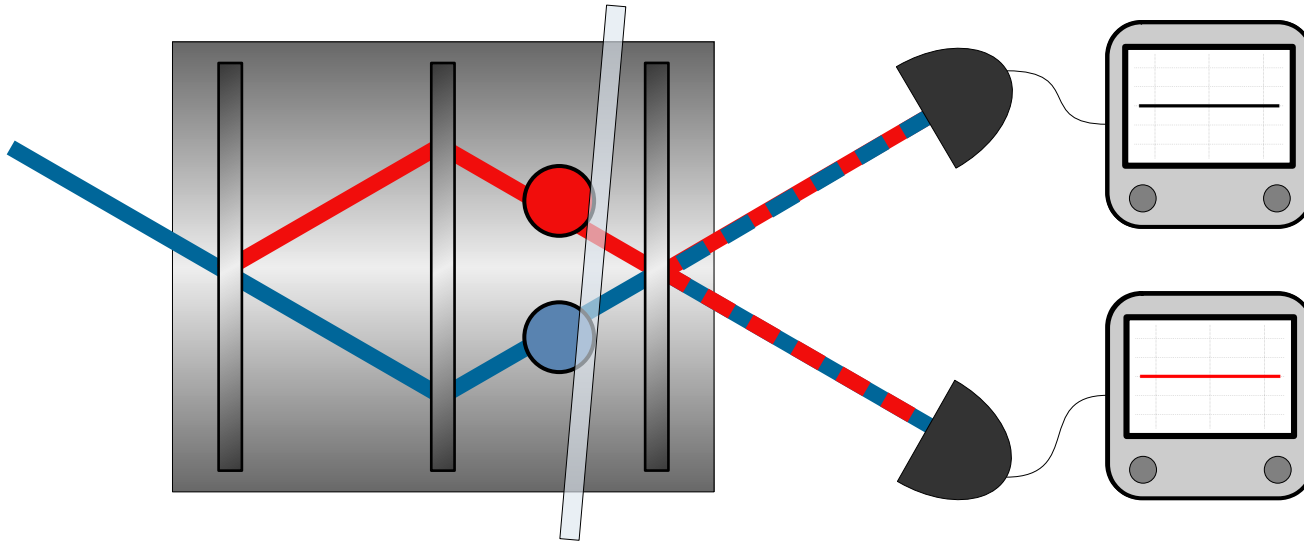
Initial state:

$$|\psi_{\text{in}}\rangle = \cos\left(\frac{\theta}{2}\right)|1\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|2\rangle$$

Path observable:

$$\hat{\Pi}_1 = |1\rangle\langle 1| \quad \hat{\Pi}_2 = |2\rangle\langle 2|$$

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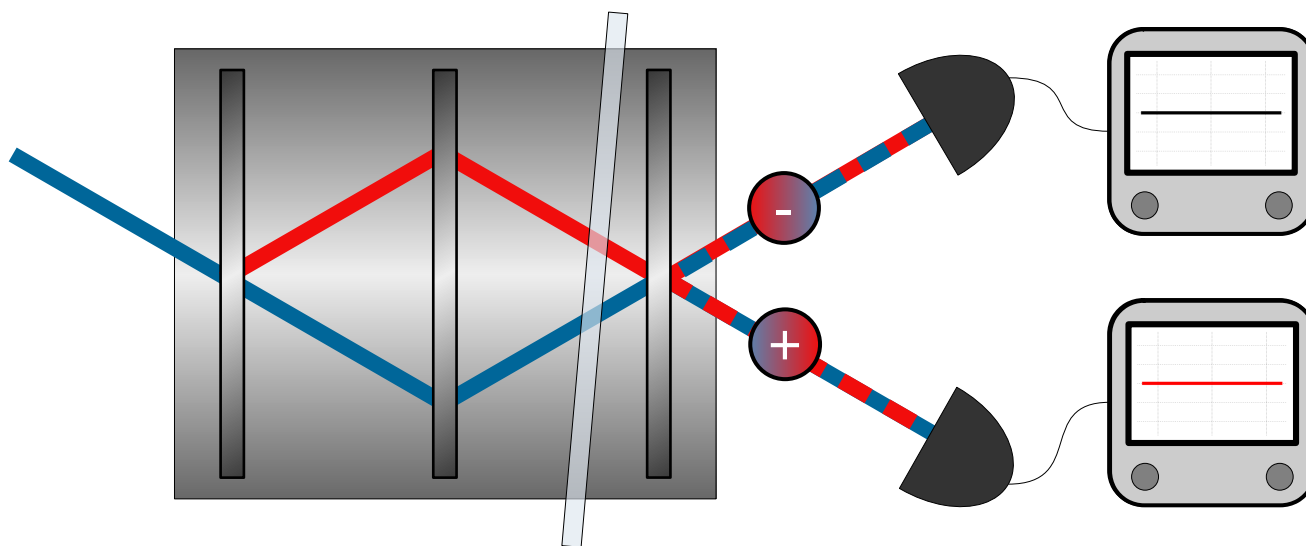
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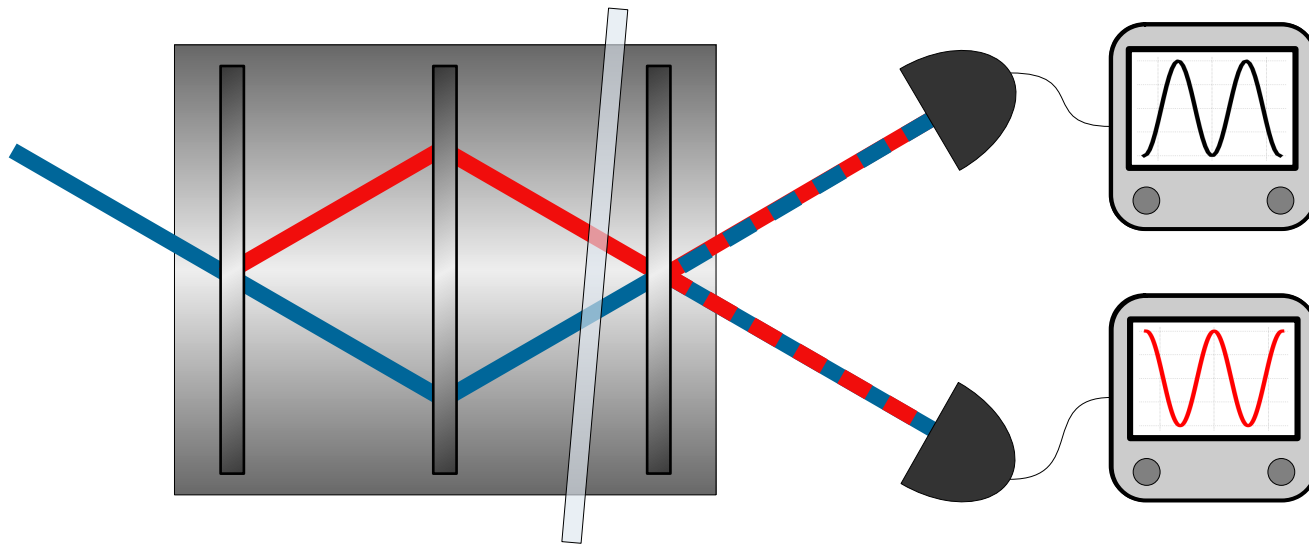
Phase shift:

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Final states:

$$|\psi_{+}\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_{-}\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

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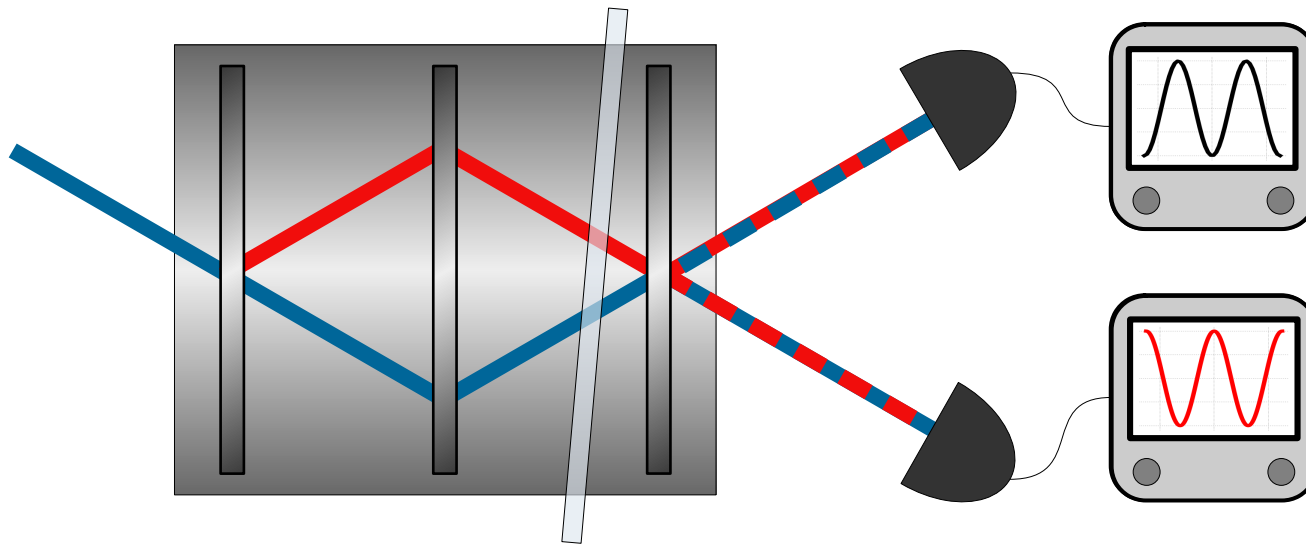
$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

Measured intensity:

$$I_{\pm}(\delta) = A \left| \langle \psi_{\pm} | e^{-i\delta\hat{\Pi}_1} | \psi_{\text{in}} \rangle \right|^2$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



Path weak-values:

$$w_{\pm,j} = \frac{\langle \psi_{\pm} | \hat{\Pi}_j | \psi_{\text{in}} \rangle}{\langle \psi_{\pm} | \psi_{\text{in}} \rangle} = w_{\pm,j}^{\text{R}} + i w_{\pm,j}^{\text{I}}$$

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Phase shift:

$$|\psi_{\text{in}}\rangle \rightarrow e^{-i\delta\hat{\Pi}_1} |\psi_{\text{in}}\rangle$$

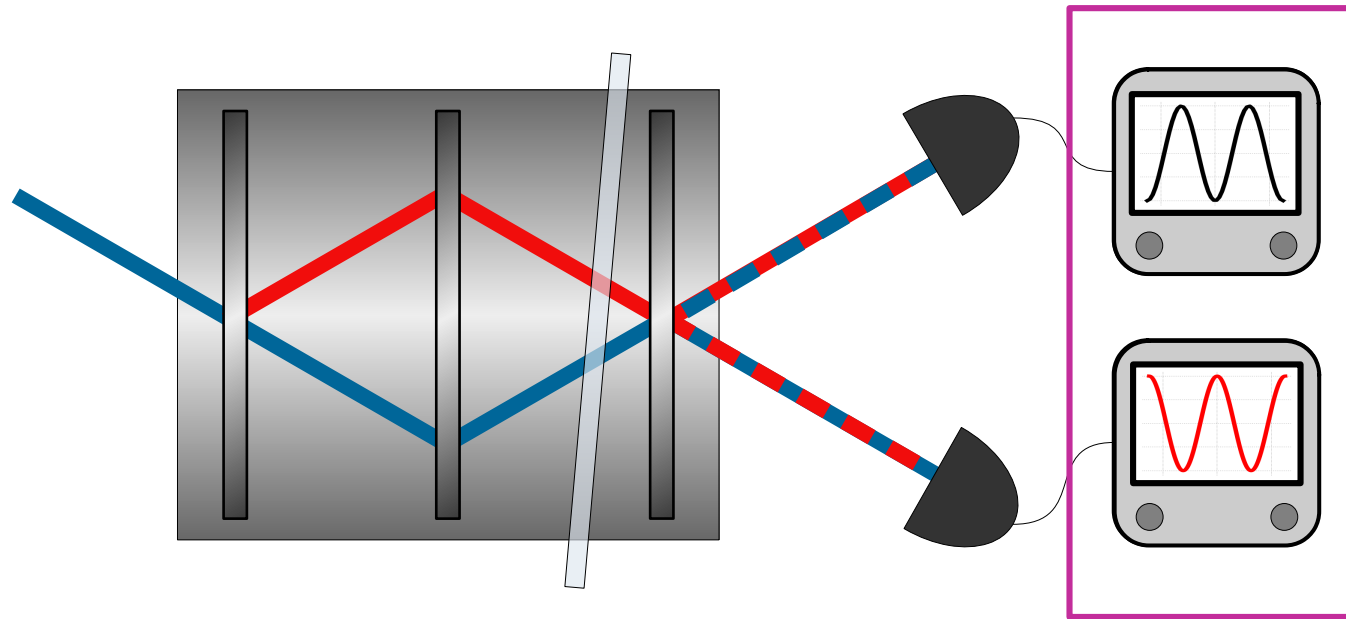
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Generalized neutron Mach-Zehnder interferometer:



Weak values can be
directly extracted from
the output intensities!

Path weak-values:

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Measured intensity:

$$\begin{aligned} I_{\pm}(\delta) &= A \left| \langle \psi_{\pm} | e^{-i\delta \hat{\Pi}_1} | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_{\pm} | \left[1 + (e^{-i\delta} - 1) \hat{\Pi}_1 \right] | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \left[1 + (e^{-i\delta} - 1) w_{\pm,1} \right] \right|^2 \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \right|^2 \left[1 + 2 (|w_{\pm,1}|^2 - w_{\pm,1}^{\text{R}}) (1 - \cos \delta) + 2 w_{\pm,1}^{\text{I}} \sin \delta \right] \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \right|^2 \left[1 + 2 (|w_{\pm,2}|^2 - w_{\pm,2}^{\text{R}}) (1 - \cos \delta) - 2 w_{\pm,2}^{\text{I}} \sin \delta \right] , \end{aligned}$$

Measured intensity:

$$\begin{aligned} I_{\pm}(\delta) &= A \left| \langle \psi_{\pm} | e^{-i\delta \hat{\Pi}_1} | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_{\pm} | \left[1 + (e^{-i\delta} - 1) \hat{\Pi}_1 \right] | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \left[1 + (e^{-i\delta} - 1) w_{\pm,1} \right] \right|^2 \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \right|^2 \left[1 + 2 \left(|w_{\pm,1}|^2 - w_{\pm,1}^{\text{R}} \right) (1 - \cos \delta) + 2 w_{\pm,1}^{\text{I}} \sin \delta \right] \\ &= A \left| \langle \psi_{\pm} | \psi_{\text{in}} \rangle \right|^2 \left[1 + 2 \left(|w_{\pm,2}|^2 - w_{\pm,2}^{\text{R}} \right) (1 - \cos \delta) - 2 w_{\pm,2}^{\text{I}} \sin \delta \right] , \end{aligned}$$

Amplitude squared Real part Imaginary part

Measured intensity:

$$I_{\pm}(\delta) = A |\langle \psi_{\pm} | \psi_{\text{in}} \rangle|^2 \left[1 + 2 (|w_{\pm,1}|^2 - w_{\pm,1}^{\text{R}}) (1 - \cos \delta) + 2 w_{\pm,1}^{\text{I}} \sin \delta \right]$$

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$$I_{\pm}(\delta) = A |\langle \psi_{\pm} | \psi_{\text{in}} \rangle|^2 [1 + 2 (|w_{\pm,1}|^2 - w_{\pm,1}^{\text{R}}) (1 - \cos \delta) + 2 w_{\pm,1}^{\text{I}} \sin \delta]$$

Phase

Terms extraction:

$$I_{+}(0) = A |\langle \psi_{+} | \psi_{\text{in}} \rangle|^2$$

$$\frac{I_{-}(0) - I_{+}(0)}{4I_{+}(0)} = |w_{+,1}|^2 - w_{+,1}^{\text{R}}$$

$$\frac{I_{+}(\frac{\pi}{2}) - I_{-}(\frac{\pi}{2})}{4I_{+}(0)} = w_{+,1}^{\text{I}}$$

Measured intensity:

$$I_{\pm}(\delta) = A |\langle \psi_{\pm} | \psi_{\text{in}} \rangle|^2 [1 + 2 (|w_{\pm,1}|^2 - w_{\pm,1}^{\text{R}}) (1 - \cos \delta) + 2 w_{\pm,1}^{\text{I}} \sin \delta]$$

Phase

Terms extraction:

$$I_{+}(0) = A |\langle \psi_{+} | \psi_{\text{in}} \rangle|^2$$

$$\frac{I_{-}(0) - I_{+}(0)}{4I_{+}(0)} = |w_{+,1}|^2 - w_{+,1}^{\text{R}}$$

$$\frac{I_{+}(\frac{\pi}{2}) - I_{-}(\frac{\pi}{2})}{4I_{+}(0)} = w_{+,1}^{\text{I}} \quad \checkmark$$

Real part extraction:

$$\frac{I_{-}(0) - I_{+}(0)}{4I_{+}(0)} = |w_{+,1}|^2 - w_{+,1}^{\text{R}} = (w_{+,1}^{\text{R}})^2 + (w_{+,1}^{\text{I}})^2 - w_{+,1}^{\text{R}}$$

Real part extraction:

$$\frac{I_-(0) - I_+(0)}{4I_+(0)} = |w_{+,1}|^2 - w_{+,1}^R = (w_{+,1}^R)^2 + (w_{+,1}^I)^2 - w_{+,1}^R$$

Method 1:

$$w_{+,1}^R = \frac{1}{2} \pm \frac{1}{2} \sqrt{\frac{I_-(0)}{I_+(0)} - \left(2 w_{+,j}^I\right)^2}$$

✗ Ambiguity in the solution

✓ Reveal nonclassicality

Real part extraction:

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✗ Ambiguity in the solution

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Method 2:

$$w_{+,j}^R = |w_{+,1}|^2 + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

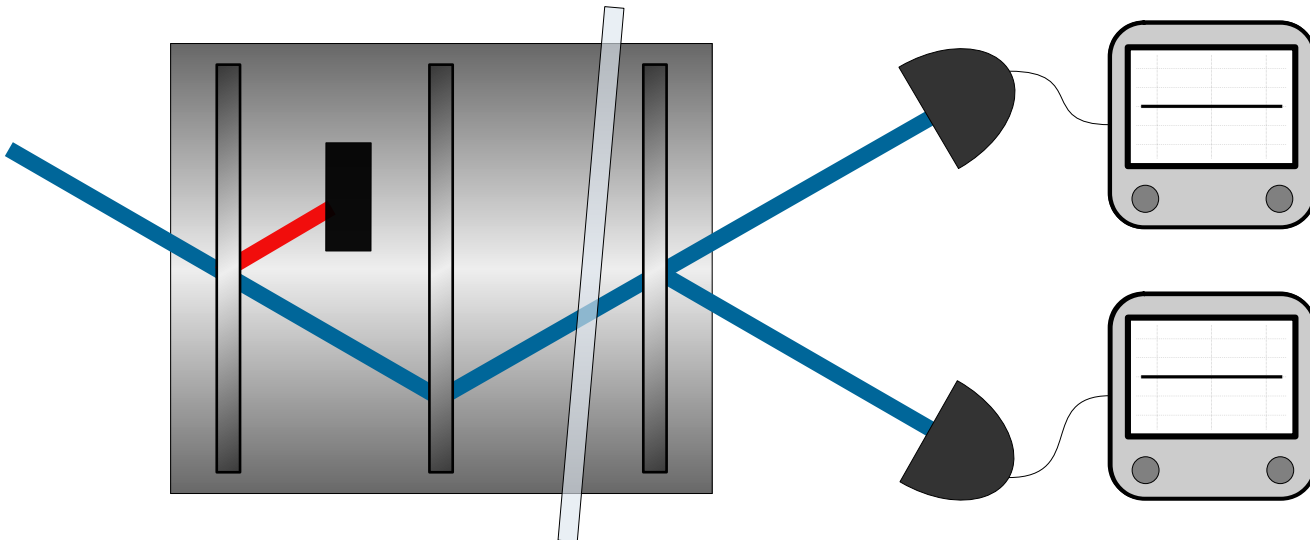
✗ Extra measurement

✓ Exact solution

Method 2:

$$w_{+,j}^R = |w_{+,1}|^2 + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

Path 2 blocked:



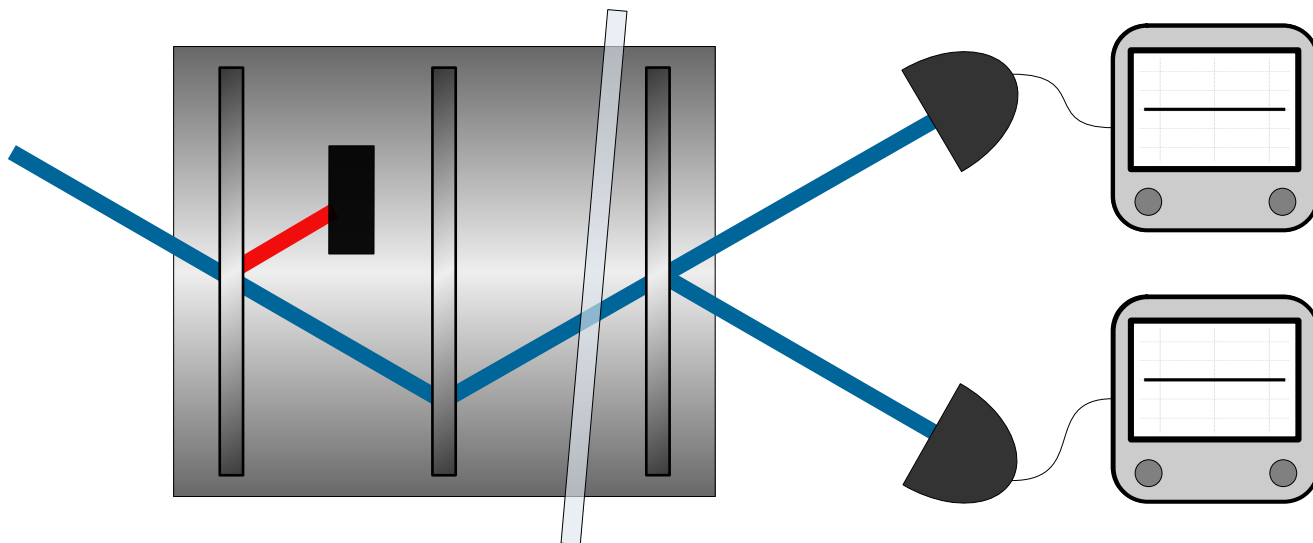
Measured intensity:

$$\begin{aligned} I_{+,1} &= A \left| \langle \psi_+ | \hat{\Pi}_1 | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_+ | \psi_{\text{in}} \rangle \right|^2 |w_{+,1}|^2 \\ &= I_+(0) |w_{+,1}|^2 \end{aligned}$$

Method 2:

$$w_{+,j}^{\text{R}} = \frac{I_{+,j}}{I_{+}(0)} + \frac{I_{+}(0) - I_{-}(0)}{4I_{+}(0)}$$

Path 2 blocked:

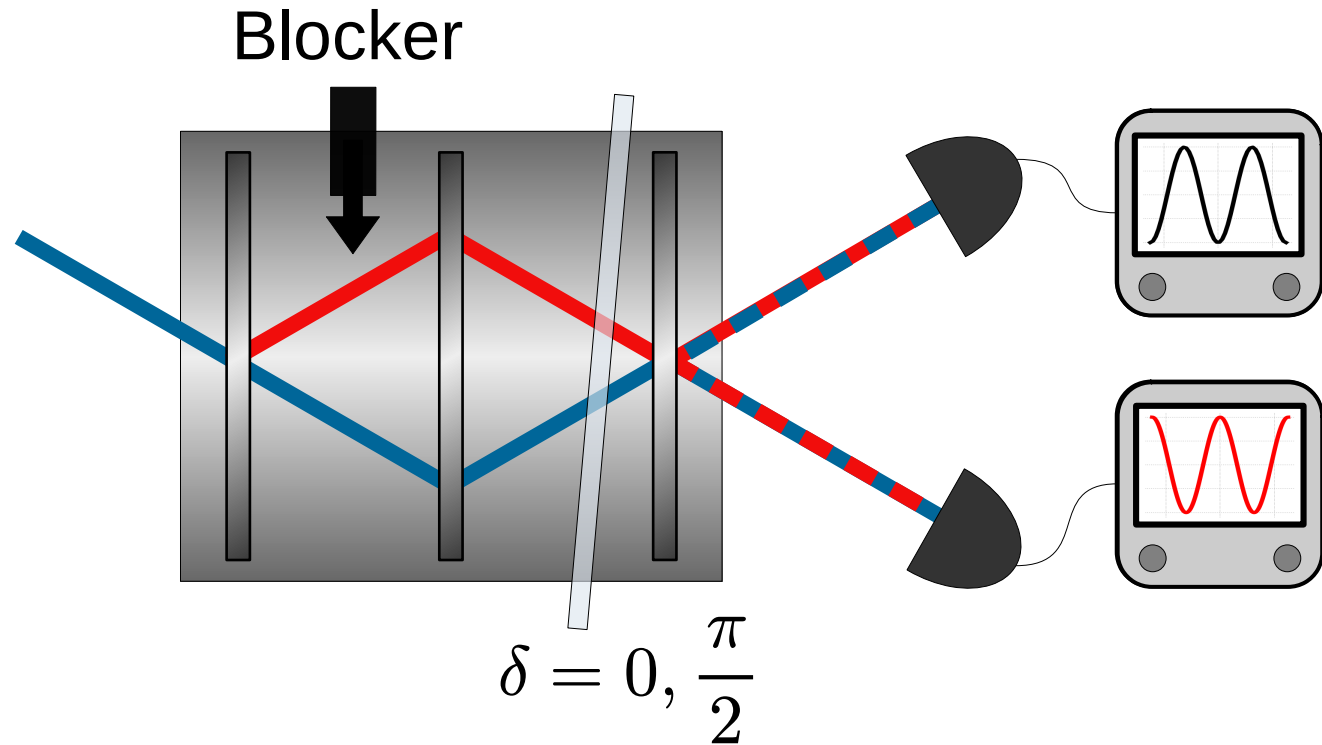


Measured intensity:

$$\begin{aligned} I_{+,1} &= A \left| \langle \psi_{+} | \hat{\Pi}_1 | \psi_{\text{in}} \rangle \right|^2 \\ &= A \left| \langle \psi_{+} | \psi_{\text{in}} \rangle \right|^2 |w_{+,1}|^2 \\ &= I_{+}(0) |w_{+,1}|^2 \end{aligned}$$

Required setup:

- Interferometer
- Path blocker
- Phase shifter



Measurements:

- No meter state and no weak interaction
- 2 measurements needed to test anomalous features
- 3 measurements needed to completely determine the path weak-values

Weak values from output intensities

